A manual to describe the construction of a lightweight strip-built sea kayak

The Vember and Vembex Build Manual v.33, February 2023

www.cnckayaks.com

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The Vember and Vembex Build Manual (red print for Vembex, blue for cove and bead only)



Design criteria

Vember is a lightweight day-tripping Kayak with a strip-built hull and a plywood deck for lightness and simplicity. Her round bilged hull gives smooth and progressive stability from upright to edged so that she is capable and reassuring in rough seas and strong winds. Strong rocker in the front half of the keel, combined with her light weight, results in a manoeuvrable and responsive kayak that is a delight to paddle.(Vembex is the lengthened expedition version, and specific details for Vembex are printed in red in this manual) We've long been interested in strip-building a round-bilged sea kayak, but had been put off by the many hours this seemed to require, in comparison with the 100 hours to complete a stitch-and-glue Shrike. To reduce the construction time, Vember combines a strip-built hull with the simple, light-weight and quick deck construction used in the Shrike family, and a first-time strip builder might take 150 hours.

Taking a successful chined hull, and creating a round bilge by drawing a fair curve to link the gunwales, chines and keel has proved successful in the past, and Vember draws on that technique. Here is the foot bulkhead template of Shrike, placed in Vember's foot bulkhead position:



This increase in volume would make the hull rise excessively out of the water, so this is compensated by a 9% reduction in length from that of the Shrike, with the beam remaining the same.

Vember's basic dimensions and facts:

Vembex, the expedition version

Length…… 4.86 m (15.94 feet)…… 5.346 m (17.53 feet)
Beam …… 0.546 m (21.5 inches)…… 0.546 m (21.5 inches)
Weight, with wire operated skeg….. 15.9 kg (35 pounds)…… 17.2 kg (38 pounds)
Approximate time to build……. 150 hours…… 150 hours
Approximate cost of materials in 2023……£700 (U.K pounds)……£770

Appendix 2 describes how to adjust the length and beam independently if you wish to modify the designs.

Useful reading prior to construction:

It is advisable to read this entire Build Manual, including the Builders' Tips and Frequently Asked Questions Appendices before starting to build. The Builders' Tips Appendix expands on certain points and gives further options which you may wish to consider. They will not fully detail the range of the various techniques of strip-building a sea kayak hull. There is a wide choice of materials, strip layout and cutting techniques and this wide choice can be daunting for the first-time strip builder. This Build Manual will describe the particular choices we made, and it should be sufficient to enable the hull to be constructed. Those interested in exploring other options will find much information in books and on-line videos. In particular, the definitive and excellent book by Nick Schade "The Strip-Built Sea Kayak ", ISBN 0-07-057989-X is highly recommended.

Materials we use:

220 metres of 19mm x 6mm timber (250 m for Vembex). Cedar is standard, but on the prototype we used paulownia for its light weight. This can be machined as cove and bead, or the square edges can be planed to fit as required. We now just use square edge strips 4.5mm thick.
One sheet of 3mm marine plywood, 2440 mm x 1220 mm.
Three to six litres of low viscosity epoxy with slow cure hardener. We use MAS brand. The quantity will vary with weight of cloth you use, and the care with which you minimise the use of the resin.
160 grams/sq. metre (4.7 oz./sq.yd) glass cloth to cover the hull, inside and out, and the deck if so desired. (Twill cloth drapes very easily and requires no cutting to conform to the shape of the hull.) We usually order 12 metres x 1 metre from East Coast fibreglass Supplies, U.K, and cut strips to make flexible tape.
Waterproof woodworking glue between the strips. (We use Everbuild 502 or Titebond 2)
Two sheets of 12mm (1/2 inch) low-cost plywood or particle board for the forms.
Polyurethane varnish or paint. CA glue(super-glue) with spray accelerator.
Kari-Tek skeg slider (aka glide box), wire and outer plastic tube, and type 618 brass end fitting for the skeg box end, or a complete Kayaksport skeg system.

Tools:

Steel metric measuring tape, at least 6m (18feet) long,
Steel metric rulers, one 1m long, one about 300mm long
Small block plane with sharpening devices
Electric jig saw with fine wood-cutting blades or fine panel hand saw
Electric drill with drill bits and screw driver bits
Spirit levels, one at least 600mm long, and one about 150 mm
Utility knife (e.g. Stanley knife) with many spare blades, adjustable sliding bevel
At least five long kayak securing straps for bending the decks, fine-bladed pull saw
Cutting board for butt jointed strips, constructed as described later
4" (100mm) gloss paint disposable foam rollers
Hand staple gun and Arrow T50 10mm (3/8th inch) stainless staples
Optional tools:
Hot glue gun with wood adhesive sticks.
Vacuum cleaner for cleaning inside the hull. Electric band saw for tapering strips

How to obtain the plans

The free plans to produce the temporary forms are available from our download tab at cnckayaks.com and can be printed by four methods, depending on what printer or cutting equipment is available:

1. If, like us, you only have access to a home A4/letter size printer, then we provide PDF files to enable the forms to be created. This is very low cost, potentially the least accurate, and the most time-consuming option. There is one sheet of paper for each of the 14 hull forms and the three bulkheads. Because of their size (greater than A4/letter), the bow and stern forms are created from offsets, as detailed later.
2. If you have access to a printer for printing at least 610mm wide (A1 size will easily suffice), then the individual forms can be printed full-size on separate sheets of paper, and then transferred to plywood. This is low cost, potentially accurate, but it takes time to cut the perimeter of all the forms.
3. The full-size plans can be produced on a single rolled sheet from any commercial printer able to print at least 841 mm wide. The continuous roll download is a 25% scale PDF that must be printed at four times magnification by the printer. We use Servicepoint in the U.K, and details are in the download.
4. If you have access to a CNC cutter, then we provide .DXF files for all the forms. This method facilitates accuracy and saves time. However, it is usually the most expensive.

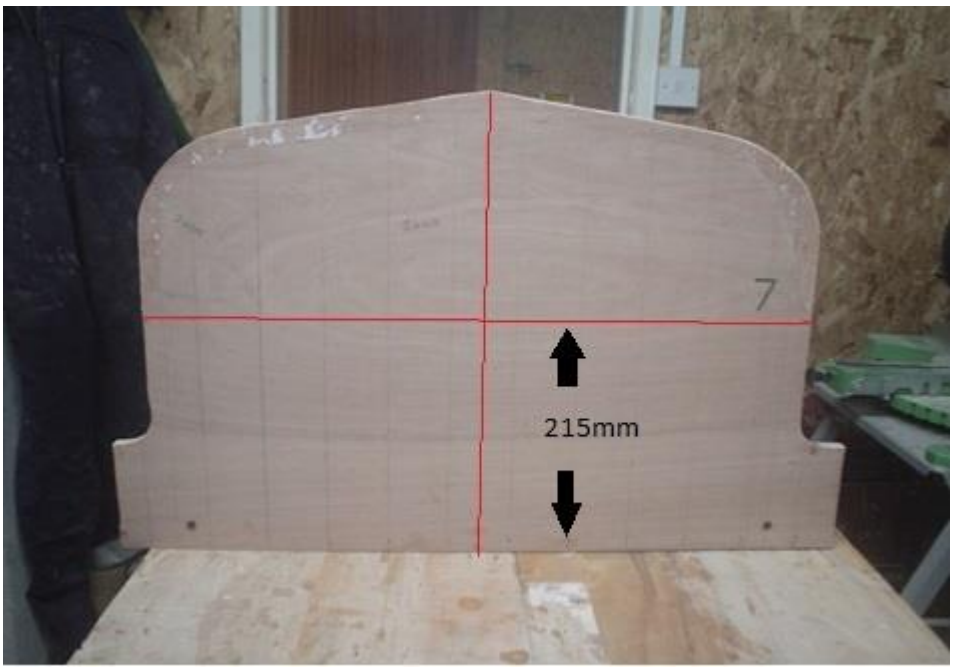
Preparing the form structure

The first and most critical stage of the construction is to prepare a flat worktop and to secure to it the temporary forms around which the hull is shaped. We used a pair of trestles, and laid one section of an aluminium ladder across them as a rigid base for a worktop. Our worktop was 4880mm x 610mm (approx. 16 ft. x 2 ft.), prepared from a sheet of 19mm (3/4 inch) shuttering plywood ripped down the centre. This worktop should be as flat, straight and level as you can make it. Care taken here will make aligning the forms far less time-consuming. Here is what you will be working towards:

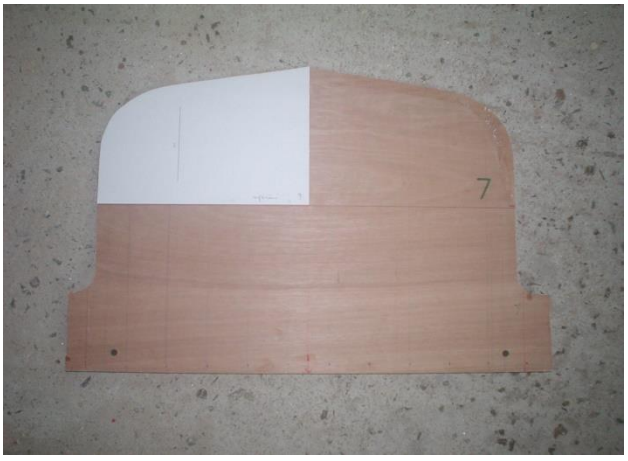


Creating the forms when using an A4/letter printer

If you are using a CNC machine or a wide printer then you can skip the next few pages, but here's how to create each form from the one A4/letter sheet of paper. Taking form 7, for example, here's the final shape:



We used a piece cut from a sheet of 12 mm (1/2 inch) low grade plywood, preserving the factory cut edge for the base of the form, which is the same width as the worktop, i.e. half a sheet width. This edge later rests on the worktop. We drew a vertical red centre-line down the plywood, and a horizontal red line representing the position of the gunwales at that station. The height of the gunwale line above the bench (datum) is printed on each sheet. For form 7 the line is 215 mm above the worktop. The curved profile is obtained by cutting out or pricking through the paper template, and then cutting the plywood to shape. The paper template is then turned over to form the other half of the hull shape. From the gunwales the cutting lines drop vertically to 100mm from the base, where they turn horizontally, as shown:

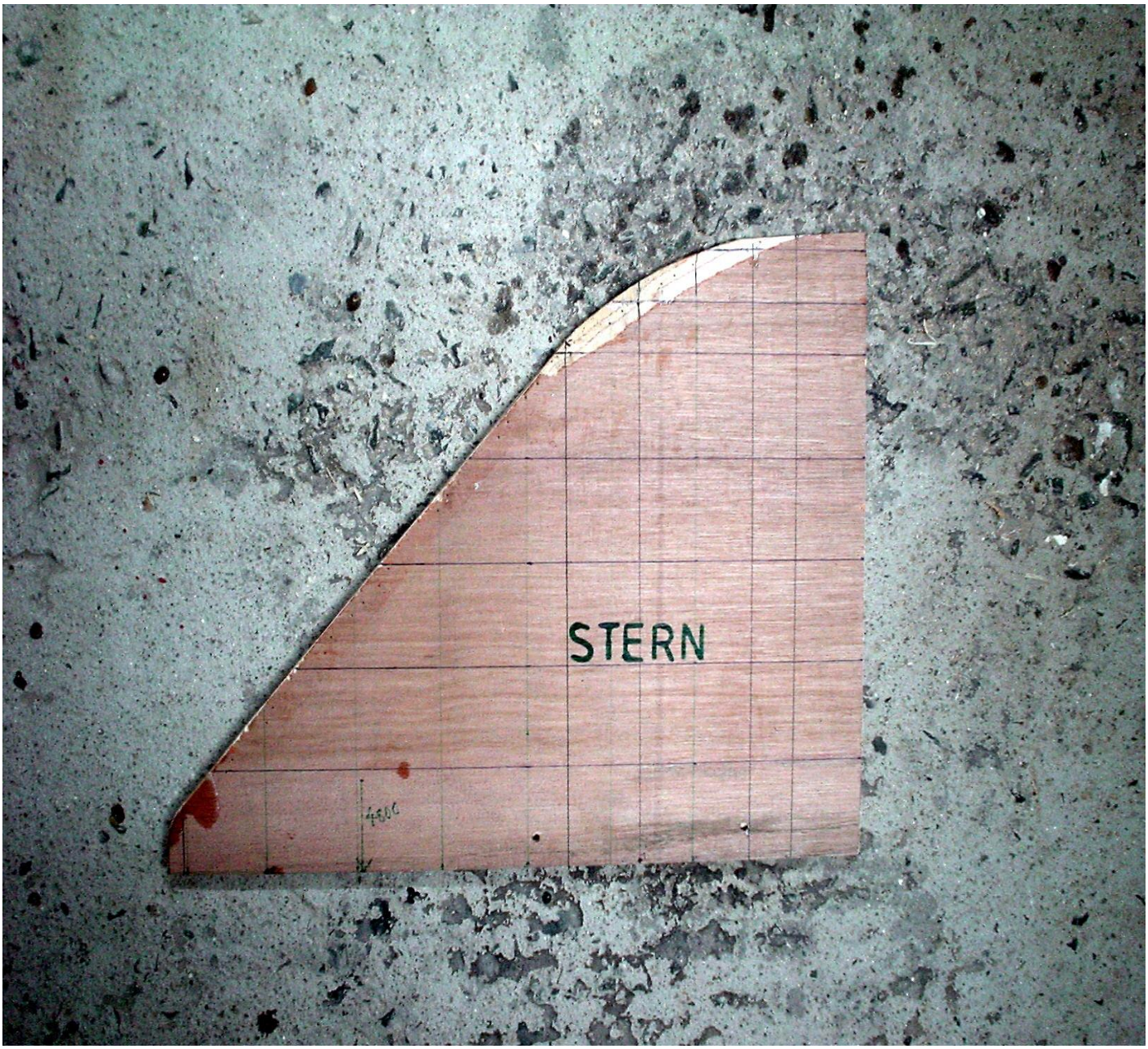




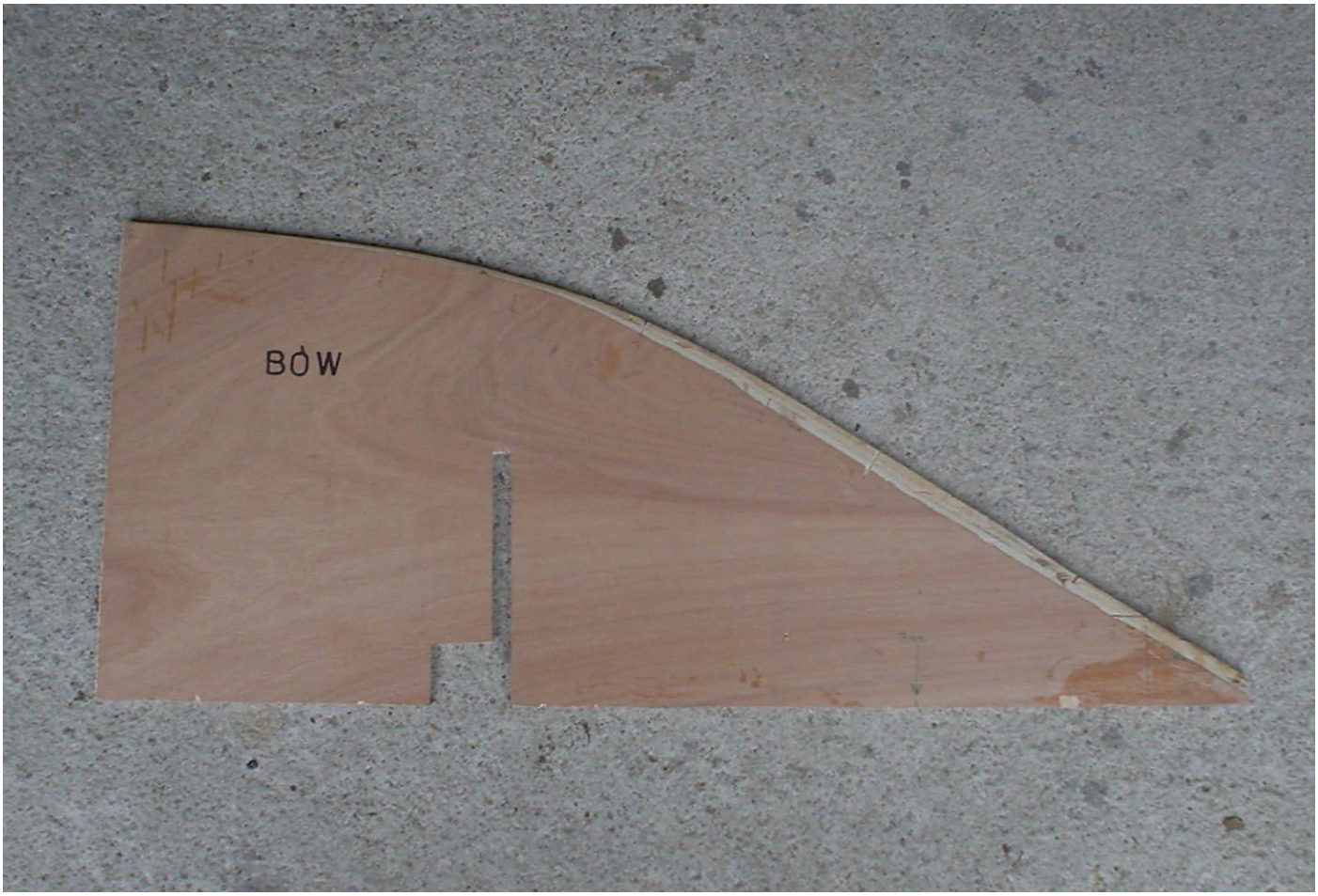
An accurately machined square timber batten, about 50mm x 50mm, is then attached to the base of each of the fourteen forms.

The stern form from an A4/letter printer (For Vembex the x-axis offsets have been increased by 10%)

The stern form is created by drawing the outline directly onto the plywood using offset (x,y) point measurements in millimetres. The x measurements are along the base, and the y measurements are the vertical heights at each x point. Taking the bottom left hand corner of the form in the photo below as (0,0), the eight offsets are:
Vember: (0,0) (10,35) (238,255) (280,285) (313,299) (347,306) (373,307) (416,309)
Vembex: (0,0) (11,35) (262,255) (308,285) (344,299) (382,306) (410,307) (458,309)
For clarity, these points are shown in the 50% scale paper print of the stern form in the A4/letter download. The points are joined with a fair curve at each end with a long straight stretch in between, as in the photo:
(Note the small area where the edge is chamfered down (bevelled) to about 2mm thick to accommodate the ends of the strips. The complete stern form is 416 mm long x 309 mm high (458 x 309 for Vembex)



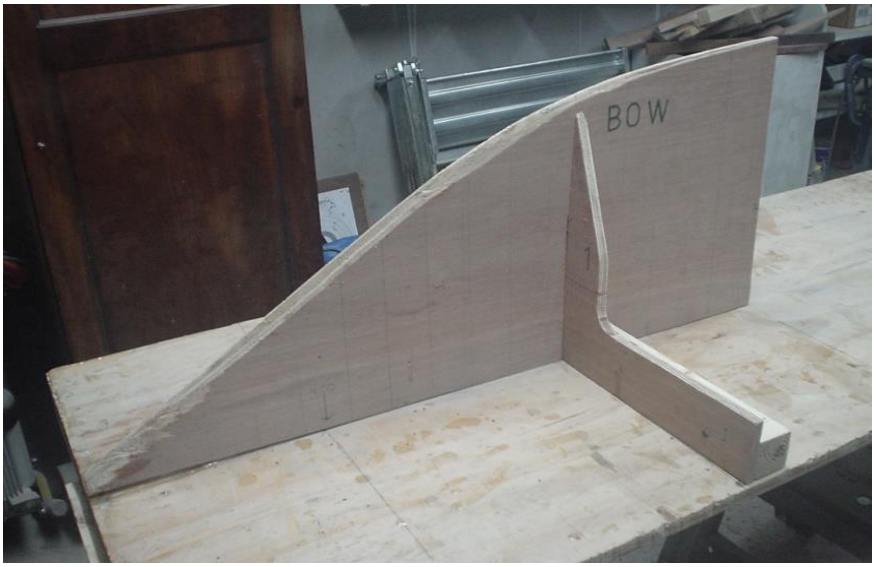
The bow form from an A4/letter printer **(For Vembex the x-axis offsets have been increased by 10%)**



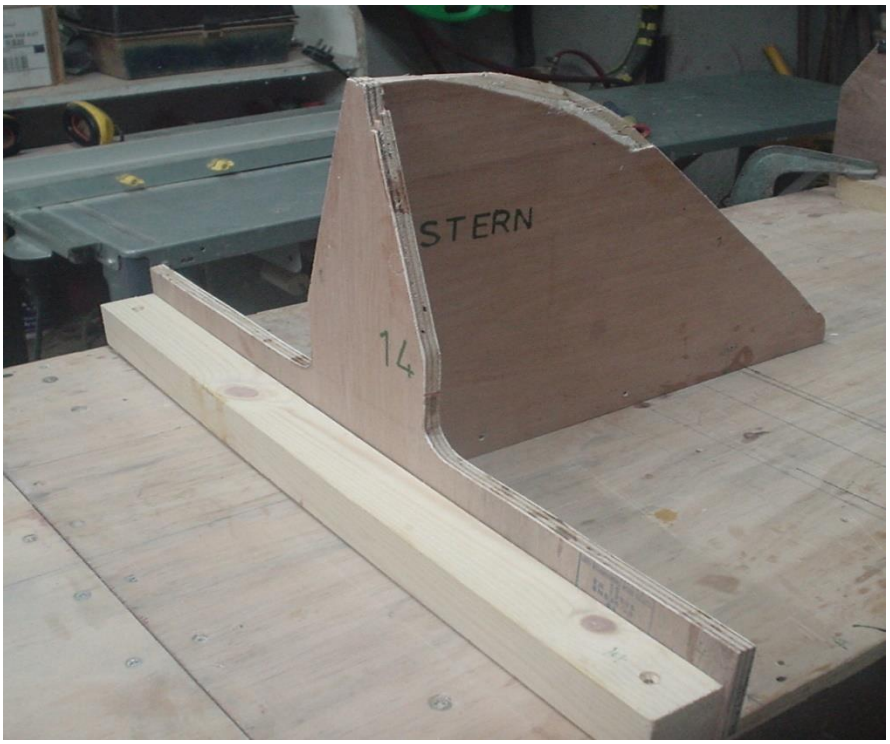
The bow form, including the slot for form 1, is drawn by using (x,y) offsets in millimetres, with the bottom left hand corner of the plywood in the photo as (0,0):
Vember offsets are:
(0,0) (245,0) (245,55) (285,55) (285,187) (300,187) (300,0) (846,0) (846,20) (469,250) (365,300) (278,326) (100,351) (0,358) (0,0)
Vembex offsets are:
(0,0) (270,0) (270,55) (314,55) (314,187) (330,187) (330,0) (931,0) (931,20) (516,250) (402,300) (306,326) (110,351) (0,358) (0,0)
These are shown in the 25% scale bow form A4/letter paper prints in the download. Check that the final size is 846 mm long x 358 mm high (931 x 358 for Vembex).
The form 1 slot in the bow form is 15mm wide and 187 mm high, with a 55 x 55 cut-out for the form base, as shown full-size and separately in the download.
(Note the area where the edge is chamfered down to about 2mm to accommodate the ends of the strips. This area extends from the stem, to just above the slot for form 1. The chamfered area on both the bow and stern forms will probably need to be extended and adjusted to make an accurate landing for your strips)

Assembling the forms on the worktop

The 14 forms are now screwed to the worktop at 300mm intervals (330 for Vembex), with the front of the bow form flush with the front of the worktop. Once the bow form and form 1 are in position, all other forms are at 300mm intervals (330 for Vembex). Mark these along a tape measure secured at form 1, not by individually marking off 300mm (330mm), which can result in accumulated errors.
Form 1 locks into the bow form:



Form 14 butts against the stern form:



Trim the sharp pointed top of form 14 so that it is level with the stern form.
Draw a line square across the worktop at 545mm (This distance can be varied to suit the length of your workbench) from the front edge of the workbench for positioning form 1, and then draw further lines at 300mm (330mm) intervals down the worktop for the other 13 forms. The forms are now screwed into position. The base of the forms must all be horizontal across the worktop, and at the same level. It is difficult to make a worktop that is accurately flat. We use steel washers or pieces of card or wood as shims

under the ends of the 50 mm square timber battens where necessary to ensure that the battens are all horizontal, as checked with a spirit level. Also, all the form bases must be at the same height. This can be checked by running a taut string down each side. Because the forms have a thickness (12mm in my case, but this can be varied to suit your material availability), and the hull tapers towards the bow and stern, the side of the forms furthest from amidships (the point of maximum beam) must rest on the station lines. This means that the side of each form with the 50mm batten attached must point towards the widest point of the hull.



Once all the forms are vertically in position on the worktop it is vital to check the fairness of the hull. A long thin wooden batten can be used to check the fairness of the keel rocker, gunwales and intermediate points. Here the rocker is being checked and adjusted by, where necessary, using thin pieces of wood under the outer ends of the transverse 50 mm square battens at the base of the forms.



<u>It is absolutely vital that great care is taken at this stage to ensure the form structure is fair in all directions.</u> To state the obvious, an unfair form structure will create an unfair hull.

The bow and stern forms are bevelled to receive the strips, as shown in the earlier photos. The ends of these two forms can flex, so they should be secured vertically and horizontally in position by screws, short lengths of batten or metal brackets. Trim the sharp tops of forms 2 and 14 to be level with the tops of the bow and stern forms.
A thin wooden batten is then stapled below the form gunwale lines (red in the below photo) along the whole length of the kayak to act as part of the form structure. (Allow the batten to fall naturally on the stern form without forcing it to an unnatural level) The first wood strip on each side is clamped to this batten if staples are not used. We chose to use clamps rather than staples, except where extra force was necessary to secure the strips to the forms. One gunwale batten is seen here:



The forms show the designed height of the gunwales, but during construction one can lower or raise the freeboard (height of the deck above the waterline) by 1 cm (0.4”) for each 15 kg (33pounds) below or

above 80 kg (176 pounds) total load, or for different sized feet. Raise or lower the gunwale batten as required.
The next photo shows the freeboard with a 180 pound (82 kg) load, and with shoes fitting snugly at size UK 10, Euro 44/45, USA 11. (Length including neoprene shoes is 285 mm)



Starting to strip the hull

Now the form structure is nearly ready for the first planks on each gunwale. Before adding the planks apply adhesive tape on all the forms to prevent the planking adhering to the forms. (We use Duct tape. Parcel tape is too flimsy.)
If using cove and bead timber, the strips are secured “cove edge up” on the forms so as to hold the bead of carpenters’ glue that is run along each cove to secure it to the adjacent bead.
Excess glue will run down inside and outside the hull. Make sure to remove this with a damp cloth before it has started to set. This will particularly minimise the effort needed later to clean the inside of the hull. Fit the strips on alternate sides, and ensure that the ends of the strips at bow and stern stay at the same height on both sides. Otherwise, cutting the mitre joints where the strips meet at bow and stern will be much more difficult. Although we chose to use clamps rather than staples, we still found it necessary to use temporary staples to secure the ends of the strips to the bow and stern forms. Take particular care to align the strips at bow and stern so as to avoid hollows that would later require excessive filling. The notched plywood U-shapes for the clamping can be seen in the next photo. We made twenty eight of these from offcuts of plywood so that we could simultaneously fit strips on both sides of the fourteen transverse forms. These plywood shapes, combined with the clamps, provide a powerful method of holding the strips

firmly against **the station line edge** of each form. Use adhesive tape to pull the strips together if the clamps are insufficient.

It is vital throughout the stripping process that no strip is allowed to drop inwards beyond its correct alignment, forming a hollow that would require much sanding of the surrounding strips. If a strip does drop inwards, in extreme cases a section of strip will later need to be glued and stapled to the recessed area, and later planed to the correct level. In some circumstances, where the dry fitted strip tends to drop below its correct level, pieces of thin wood can be temporarily hot-glued or CA glued underneath the strip and across the gap to prevent the strip dropping inwards. This is a useful trick, especially near the ends where there is twist in the strips. CA glue (super-glue) with spray accelerator is a great invention)

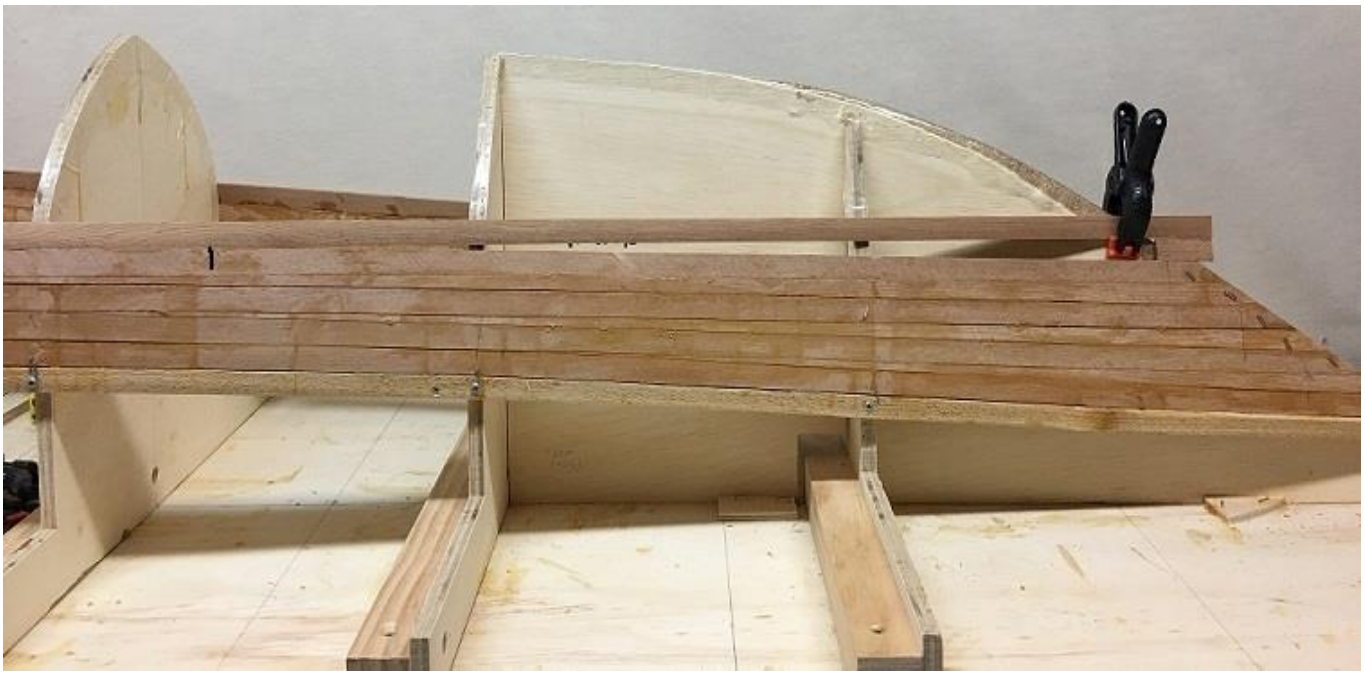
Because of the upward curve of the gunwales at bow and stern, when seen afloat, the first planks on each side are curved downwards on the upside down forms, but the planking eventually needs to meet the keel, which is a straight line. To deal with this, tapered gaps are left between the first five full planks from the gunwale, and tapered strips (splines) are later fitted in these gaps, as described below. The gaps for the tapered strips have a maximum width of one strip, 19mm.



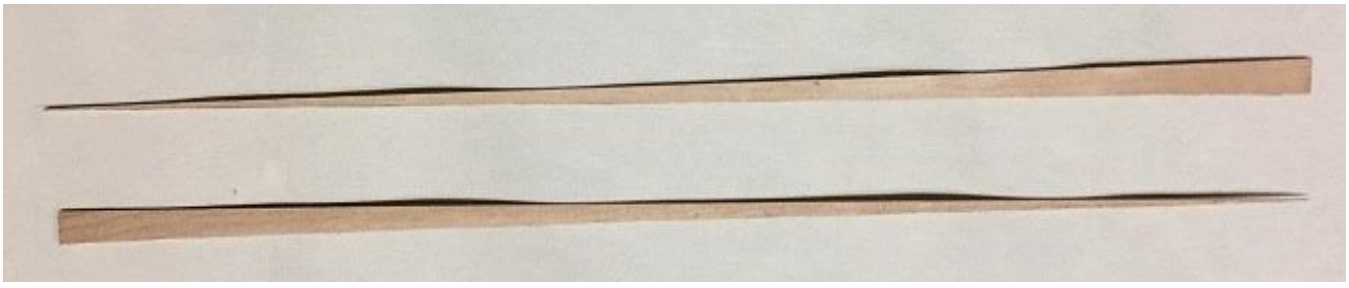
Making and fitting the tapered infill strips.

Tapered infill strips are necessary at bow and stern to enable the strips from the gunwale to meet the keel strips in a straight line. There are four tapers on each side between the first five strips from the gunwale, making a total of sixteen tapers.

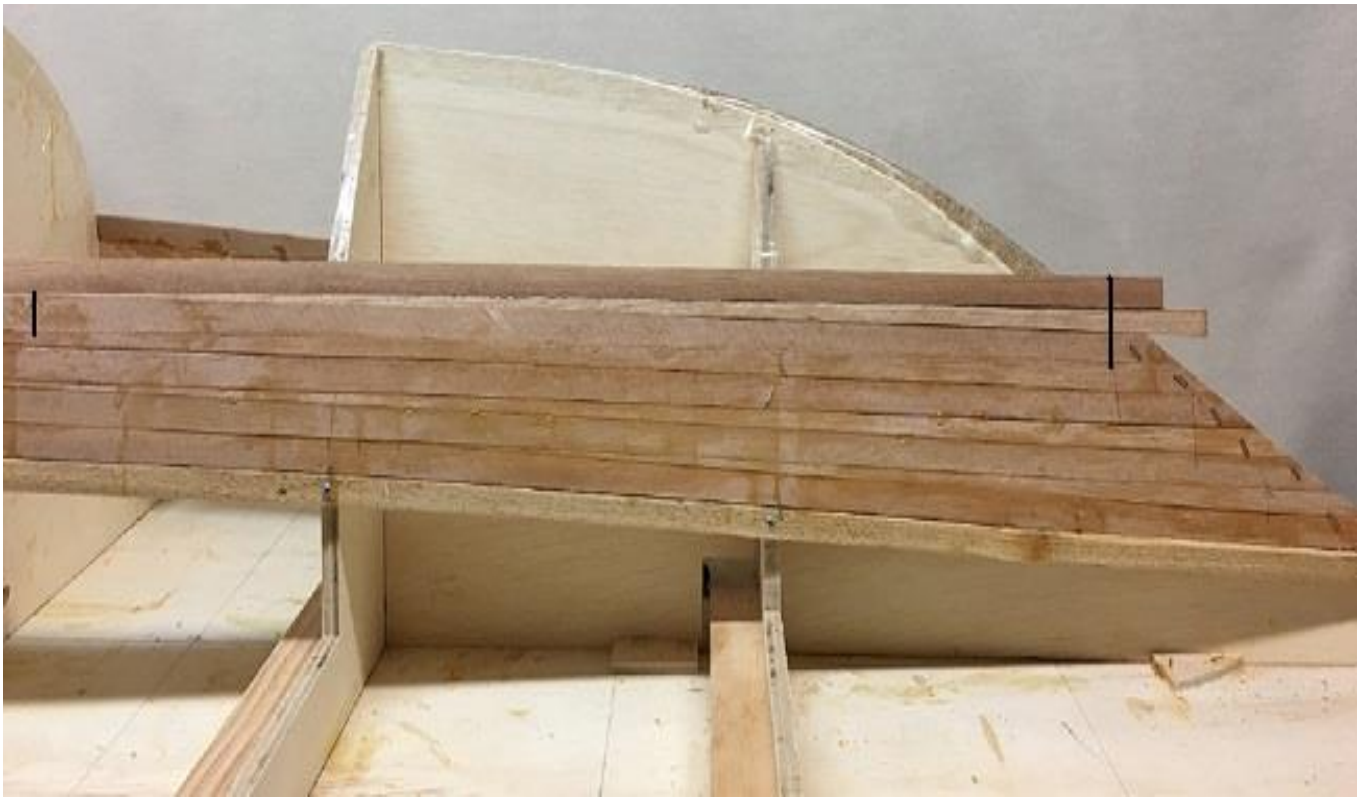
At the bow and stern the tapered strips must be no wider than a full strip, so that the wide end of the tapered strip can completely fill the gap. Temporarily clamp or staple the next strip in place, with a short (about 30 mm) length of strip temporarily filling the end gap. Let the strip take up its natural curve, and the gap should disappear about 800 to 1100 mm from the end gap. Use a soft pencil to mark the point on the lower strip where the gap disappears. Measure the distance from the end gap to the pencil mark. Call this length L, and add at least 25 mm for luck.



Take a fresh piece of strip, and cut it to length L + 25 mm. A band saw works well for this task.Draw a diagonal line on the face of this strip. Band saw along this line to produce two identical tapered strips of length L + 25, tapering smoothly from full width to zero. If the saw cut is uneven, lightly plane the surface. (If using cove and bead you will need to shape a bead on the tapered piece with the cove. The other piece will be discarded.)



Test fit one taper by sliding it into the gap until it reaches the pencil mark you made earlier. If the fit is poor, plane the taper until you are satisfied with the fit. Draw a vertical “witness mark” in pencil across the taper and the two adjacent strips to enable you to refit the strip in the correct position when you have applied the glue.



Apply glue to the whole relevant length of the top of the lower strip and the top of the tapered strip. Clamp the tapered strip and the top strip in place. Ensure that the tapered end stays flush with the adjacent strips. Ensure that the tapered strips are kept at the same level as the surrounding strips. We've used partially-inserted temporary staples across the thin end, with the legs of the staple in the strips above and below, like a football (soccer) goal post. It is all too easy to recess the thin end of the splines, thereby necessitating much filling later on in the construction. If all else fails, a finger behind the spline for a couple of minutes will help. Temporarily staple the ends of the top strip and the spline to the bow or stern form. Clean off the excess glue.

(If using cove and bead strips, the coves from gunwale and keel are arranged in opposing directions so that where they eventually meet, infilling tapered strips only require a bead on each side. This is far simpler than attempting to reproduce coves. In addition, both keel strips will have their cove pointing outwards.)



On the Vember in the photos, there are four tapered strips each side at both bow and stern, a total of sixteen. The purpose of these is to gradually fit the later strips more parallel to the keel, so you may need to fit more tapered strips of varying width to achieve this result. Check the distance between the ends of each strip and the centre line of the keel, and gradually make these distances approximately equal by fitting tapered strips of the appropriate width. This will simplfy the fitting of the last strips near to the keel. The tapers can be cut with a craft knife or band saw and then planed smooth. They are cut with a straight taper, and the thin strips take up any necessary curve, as described earlier. (If using cove and bead, the cove side is preserved throughout. The tapered side is planed and sanded to produce a bead.) The ends of the strips at bow and stern may be mitred to meet at a point. The sharp, ugly and fragile points are later cut and planed square to the keel line to produce a landing for a bow covering strip. This covers up any imperfections in the mitres. An easier alternative to these awkward mitres is to cut the first strip flush with the far side of the form at the angle the opposite strip will take. The opposite strip will extend well past the end of the first strip, and it is later trimmed to length. All the second strips can extend out of the same side of the bow. This avoids the difficult task of keeping the strips on each side at exactly the same height, as would be required if the joints were constructed in alternate directions, similarly to when the fingers of each hand are interlocked. (Think of “cats’ cradles”, the children’s’ game.) All the strips are later trimmed back close to the form, so accuracy is not essential:



When strip-planking the hull it is much easier to use strips of less than full length. The ends of the shorter strips at bow and stern can be adjusted and re-adjusted for fit, and a simple butt joint suffices later to join the two lengths of strip where they overlap. Butt joints should be staggered from strip to strip to avoid a possible line of weakness. T he main function of the strips (which could be made of foam) is to separate the two layers of glass so that the benefits of the composite material are obtained. (Skin further from an applied force is under tension, and glass fibres are very strong under tension) We apply a light clamp across the butt joint while the glue sets. Butt joints are made between forms. A joint on a form is harder to clamp. When you fit the next strip a light clamp across the new strip and the butt joint below will keep the curve smooth while the glue sets.

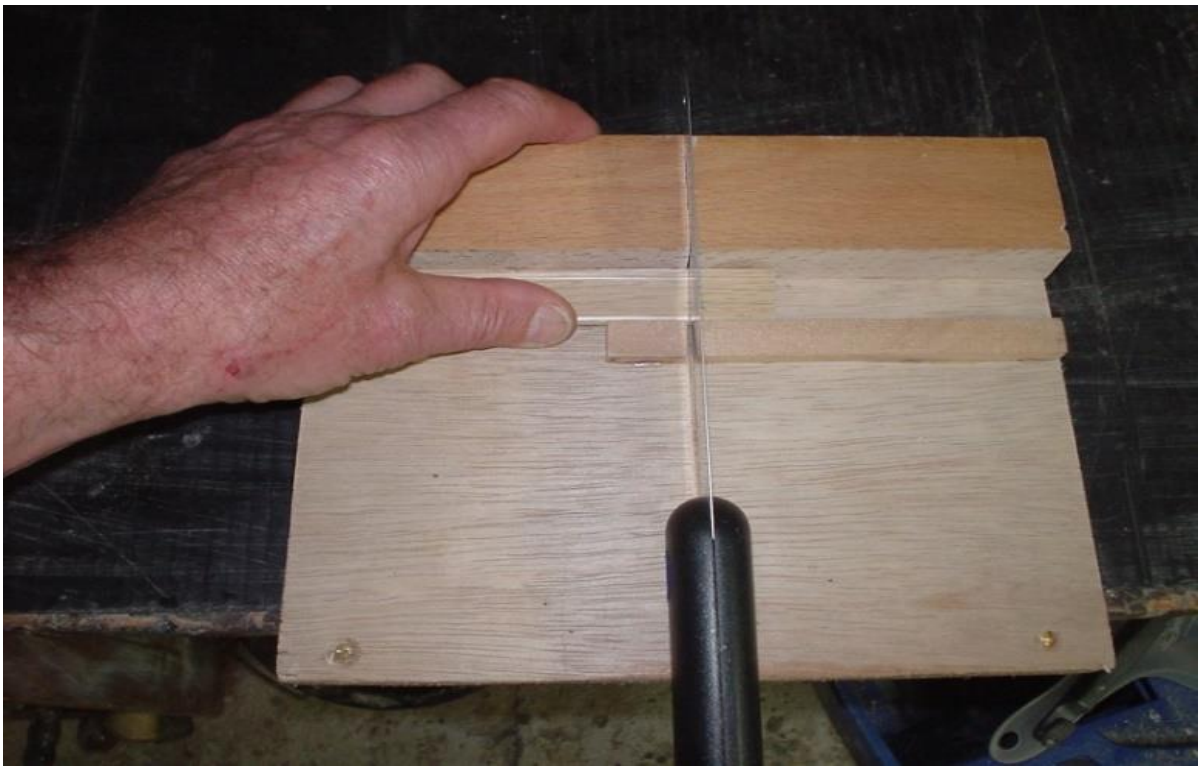
It is vital at this stage to accept that an accumulation of small errors in drawing, cutting and assembly can produce slight unfairness (irregularity) in the curve of a strip. This must be remedied, either by packing out a form with wood shavings or cardboard where a strip crosses it, or, as a last resort, moving

a form very slightly. Forget about measurements and plans. Just make your kayak beautiful and symmetrical. Time spent just looking at the hull is never wasted. Your eye is more accurate than my tape measure!

Making a strip cutting board

To cut accurate square butt joints we built a cutting board and combined this with a fine bladed pull saw:





The exciting planking now proceeds. We used the U-shaped pieces of scrap form plywood to enable each strip to be clamped, rather than stapled. This meant that we only fixed two pair of strips each day, one pair in the morning and one in the evening, but we were in no hurry. This hull is made of Pawlonia:



The clamps are later removed. If a strip pulls away from the form when the clamps are released, staples or hot glue should be used to maintain the desired shape. The hull is ready for the next pair of strips:



Continue stripping on alternate sides until the strips protrude above the stern form. Saw and plane any protruding ends of the hull strips level with the top of the stern form.
Lay a pair of half-length keel strips side-by-side down the centre of the hull from about mid-length of the hull so that they extend beyond the stern to the point where the form goes steeply down, and where eventually they will join a stern capping strip. These two keel strips should differ in length by at least 300 mm to avoid a weak spot in the keel. Bevel the meeting edges along the keel strips so that they lay over the angle of the keel forms without any gaps. Drill and temporarily nail them to the tops of the forms, and glue them together, thereby preserving the designed keel rocker shape. These nails can be seen in the next photo.
Continue stripping the hull until the strips begin to appear on the top of the bow form. Then saw and plane the top of any protruding hull strips flush with the top of the bow form so that the final pair of bevelled half-length keel strips can lay flat and unimpeded on the bow form. You can now fit and glue and temporary nail the bevelled pair of keel strips onto the front half of the hull, glued and butted to the stern keel strips.



From now on, each hull strip will have to slot into the V-shaped recess formed by the keel strip and the hull strip below. Plane the edge of the keel strip to make visible the edge joint between the keel strip and the new strip so as to visually check for an accurate fit as the planing of the hull strip continues. The outside edges of the keel strips must be given a changing bevel throughout their length, so there is a smooth transition from strips to keel, as shown in the nect photo:

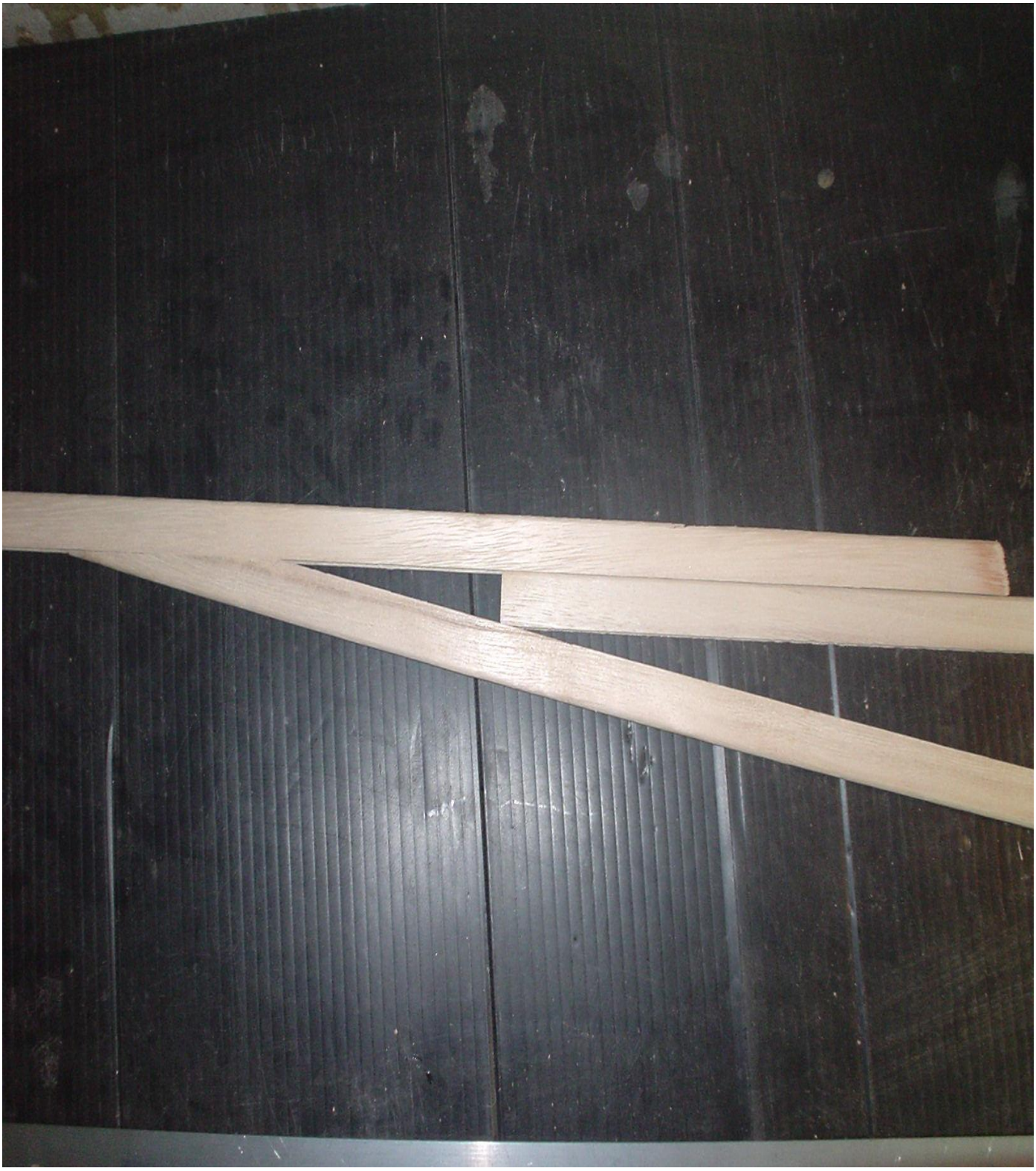


(If using cove and bead strips, the coves from gunwale and keel are arranged in opposing directions so that eventually meet, infilling tapered strips only require a bead on each side. This is far simpler than attempting to reproduce coves. In addition, both keel strips will have their cove pointing outwards.)

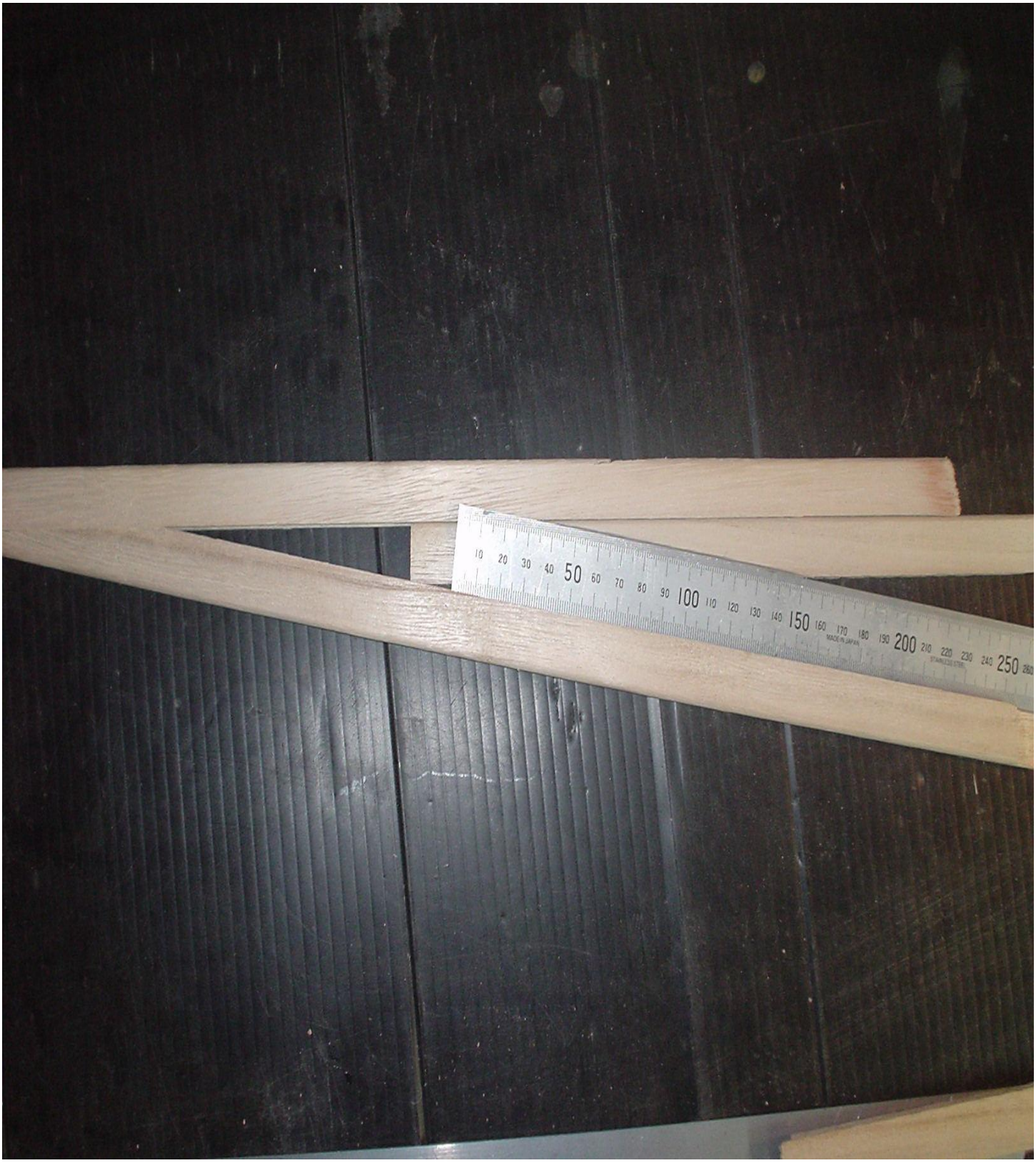
Joining the curved hull strips to the straight keel strips:
The strips are now fitted alternately up from the gunwale and down from the keel so that they interlock. Where the curved strips from the gunwale meet the straight strips from the keel, and vice versa, they are cut at the correct angle, as described below (for cove-and-bead, a bead is planed and sanded on the cut edge), and the strip is glued in position. Here's one method to ensure a good fit:
The first photo is a bench mock-up of the junction between an existing horizontal keel strip (cove down) and a gunwale strip (cove up) coming from below:



The next photo shows the end of a keel strip, (cove down, if using cove and bead), that we will fit in between the existing strips:



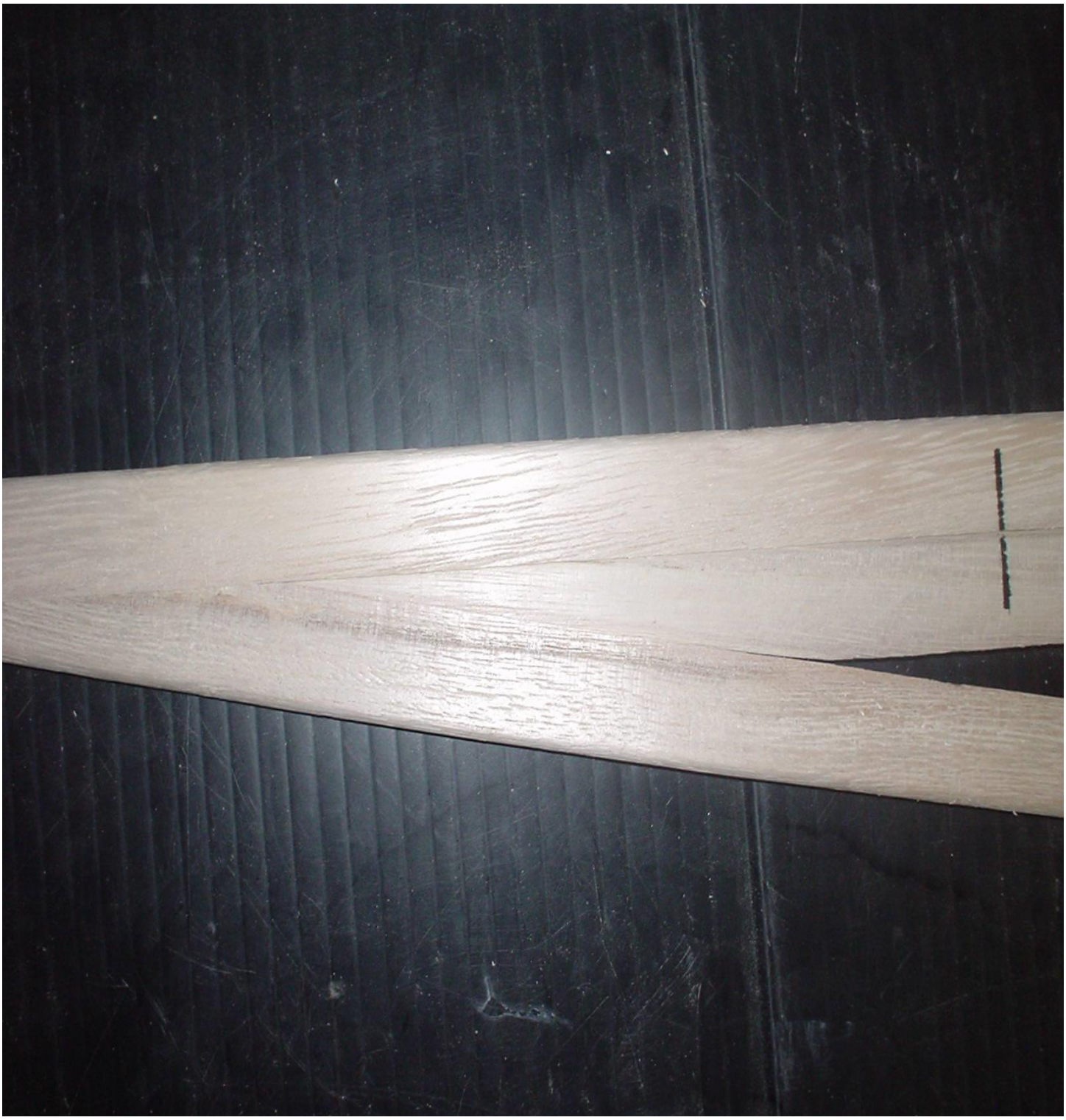
A rule is laid parallel to the lower strip and a pencil line is drawn on the new strip:



The strip is sawn and planed along the pencil line. (For cove-and-bead, a bead is formed on the cut edge with a block plane and coarse sandpaper.)



The shaped strip is inserted dry to check the fit. A “witness mark” is then drawn by soft pencil across the two adjacent strips so that the strip can be later glued into the same position:



To simplfy construction,the inserted strip is not full length. The same procedure is carried out with another strip from the other end of the kayak. Where the two strips overlap, a simple glued butt joint is marked and made using the strip- cutting board on the bench. These butt joints are staggered to avoid weak points. The two strip sections are then glued into their correct positions with the aid of the witness marks. (If the hull is to be clear-finished, these marks should be kept faint so they are easily removed with a rubber or fine sandpaper). The next strip to fit in the mock-up would be one curving from below, (cove up, if relevant. The only difference here is that any **top** edge (or cove) of the new strip would be cut off, parallel to the side of the **upper** strip, (and then formed into a bead, if relevant.).

This view from the stern shows the interlocking tapered strips. By this stage you may have sufficient confidence to fit these shorter strips in one length, using witness marks at each end.
In the next photo a pair of strips parallel to the keel is being clamped while the glue sets:



If using cove and beadstrips, when the gap between the gunwale and keel strips is less than three strips wide, any coves are planed off the next keel and gunwale strips, leaving a tapered gap less than one strip wide, with square flat inside edges.

If using rectangular strips, continue stripping until the gap is less than the width of one strip. Securing these last strips while the glue is setting requires ingenuity:

(The below photo is of a stapled Vembex hull constructed in Pawlonia by Damian in Plymouth, UK.)



Place a length of strip to just cover the final gap, and tape this securely in position. Draw around the shape from under the hull with a sharp pencil. Saw a little outside the line and plane it carefully and repeatedly until it just slides into position. Start the fit at one end and gradually work along the strip. It is then glued into position. In the below photo the strip to infill the final gap rests on the hull to the left of the gap. The strip has been (very!) roughly jig-sawed to the traced outline, and now requires careful planing so that it is a sliding fit into the gap:



The infill tapered strip is glued into position:



Bow and stern capping and fairing the hull

Use a saw and plane close to the bow and stern forms to produce a flat surface for the bow and stern capping pieces that are then glued, shaped and sanded. (It's a relief to plane off the ugly rough joints to produce a smooth outline.) The bow capping can be a piece of strip, held in place by staples or temporary screws until the glue has set. We laminated the stern capping in place from four layers of 2mm softwood, but a hull strip offcut would probably have taken this tighter curve if we had warmed it or soaked it in hot water. Now the entire hull is planed, sanded and filled to remove the edges of the strips and to fill any holes and voids. We used West 406 colloidal filler powder added to the epoxy resin to match the colour of the paulownia. For other timbers, wood dust from the sander can be used.
Here's the faired hull after planing, sanding and filling:



Glass sheathing the hull

For the remainder of the construction we used epoxy resin, taking precautions against allergic reaction to the resin and, particularly, the sanding dust. When sanding we used barrier cream on our arms and hands, disposable gloves, a disposable paper suit including a hood, and a respirator (dust mask).
Twill glass cloth conforms exceptionally easily to the complex shape of the hull, and no cutting was required at the bow and stern. (Twill cloth is woven from pairs of glass strands, rather than single strands.) To give oneself time to apply the cloth a slow setting low viscosity resin should be used, and applied on a cool day (less than 20 C, 70 F.) We poured the resin, half a cup (120 ml) at a time on to the top of the hull and spread it rapidly with a plastic squeegee (like a large credit card) held at 45 degrees to the surface to wet out the cloth. A helper is useful throughout the glassing of the inside and outside of the hull, in order to reduce the time taken, and to complete the task before the resin starts to set. The helper can mix all the batches of epoxy when required. When the last batch has been poured on, the helper can join in scraping off the excess resin. In this way the helper maintains clean gloves throughout the mixing processes.



The resin is then pulled down from the keel towards the gunwale:





Remove all excess resin with the squeegee at 90 degrees to the surface until there are **no shiny patches**. Here a credit card is being used for the final delicate stage:



Any extra resin adds weight and does not add strength. The edges of the cloth are trimmed after a few hours when the epoxy is firm, but not hard. Extra coats of resin should be rolled on to fill the weave of the cloth, preferably within 24 hours of each other so that the coats bond together. If left too long the hull should be lightly sanded before coating.
After another 24 hours it is a good plan to fit a fibreglass tape keel strip to help protect the hull from scraping on rocks. Three parallel strips of 50 mm wide good quality masking tape are run down the keel and around the bow and stern, ensuring that the first strip, down the central ridge of the keel, is straight. The other two tapes are placed either side of the central tape, with a 2 mm gap between them and the central tape. The central tape is then removed. This seems a complicated way of achieving a straight keel strip, but it works. 50 mm wide glass tape is then epoxied into position between the two parallel tapes, first brushing on the minimum resin to bond the tape and then to remove any white areas.
The masking tape should be removed after a couple of hours, when the resin will not run down the hull, but the masking tape will come away cleanly.



Turning the hull over

After a few days (five days at 20 degrees C, 70 F, to achieve full hardness) the hull is released from the forms, turned over, and placed on a pair of padded cradles shaped for the hull. These cradles can be prepared while the hull is upside down and curing. Also, this is a good time to make the masik (the curved foredeck beam) and prepare the skeg system, as described later in this manual.



When the hull has been turned over, immediately use a temporary wooden strut to preserve the beam at 546 mm.

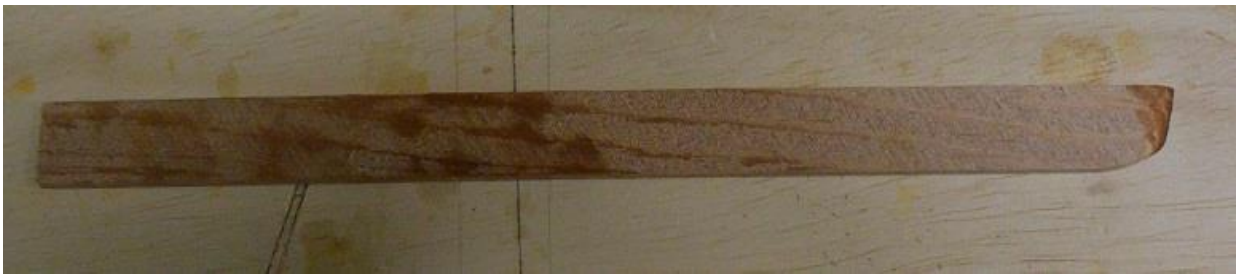


The sheer clamps, (thin timbers that will line the inside of the full length of the inside of the gunwales) can also be temporarily clamped to the outside of the gunwales to preserve the shape. Note that the temporary strut shown below has notches that are sized to accommodate later the sheer clamps. The

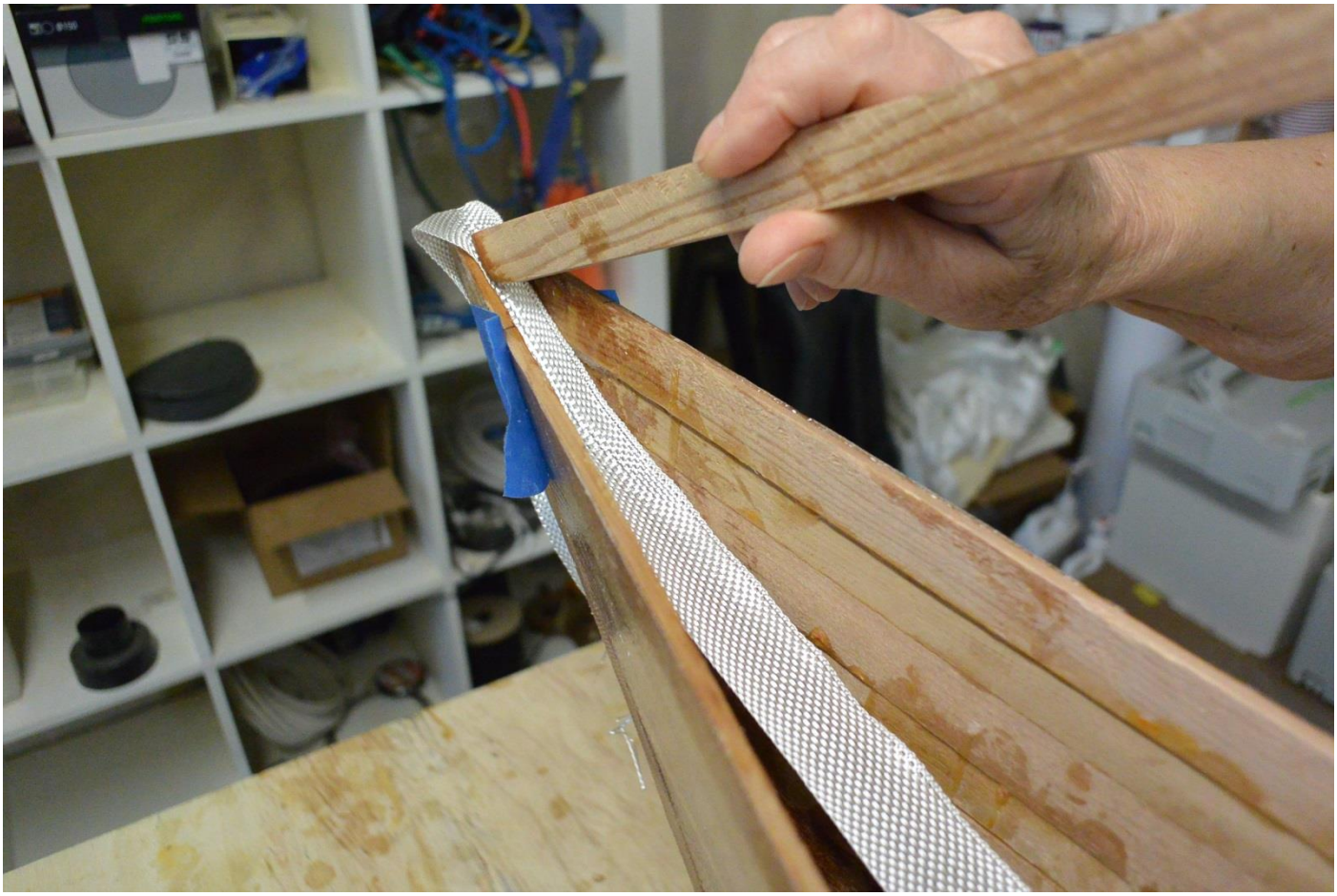
wedges shown reduce the effective size of the notches to the thickness of the hull before the sheer clamps are fitted.



When first turned over, the bow and stern are particularly fragile, and thickened epoxy should be applied to these areas, followed a few hours later by a strip of 25 mm wide tape. This is a tricky task. Begin by cutting a piece of tape to cover the area shown in the photo, and folding a crease down the centre of the tape. This won't be very effective, but it's worth trying. Now prepare a piece of your strip timber with a curved end to serve as your surgical instrument:



Lightly brush a very thin coat of epoxy resin down the area to which the tape will be applied. Use masking tape to secure one end of the glass tape over the end of the kayak. With one hand, hold the other end of the glass tape up and away from the wet resin, and, with the other hand, very carefully press the rounded end of the stick onto the centre of the glass at the top of the centre line of the kayak. Gradually lower the tape and press it into position. Once the tape is securely positioned, gently brush just sufficient resin over the glass tape to remove all dry spots. We describe this process in detail as without this level of care it can end in a nasty mess and much frustration.



Cleaning up the inside of the hull

The inside is scraped, planed, and sanded until the surface is sufficiently level and clean to enable a layer of glass cloth and epoxy to be applied without voids and air pockets. This is quickly said, but it can involve much effort and time if excess glue has been left inside the hull, and if the strips are uneven in thickness. The glue blobs on the flat area areas are best removed with a block plane, and on the curved area using a two-handled scraper with a curved blade:



A block plane will deal with uneven strips on flat areas, but on curved areas with very uneven strips, a Stanley Shaver tool, pulled across the strips, will rapidly and brutally remove excess wood:



The interior of the hull is than sanded with coarse sandpaper, leaving a slightly roughened surface for effective bonding of the subsequent glass cloth. Finally, any remaining voids are filled with thickened epoxy, which is left to harden for 24 hours.

Glass sheathing the inside of the hull

The inside is coated in glass cloth and epoxy **as soon as possible after the hull is turned over**, in order to prevent distortion as the hull cures. The outside of the gunwales can be masked and a curtain of polythene hung outside the gunwales in order to protect the outside of the hull from drips of resin. Again, remove all excess resin that collects in the keel area. **Minimize the shiny patches**! Time pressure to complete the task before the resin sets can be avoided by glassing in three pieces, with the middle piece applied last, and overlapping the front and rear glass by about 50 mm.



Measurements list for reference (Vembex in red)

Length overall: 4.85m (15ft 11”) (5.35m (17.53 feet))
Beam: 0.546m (21.5”) (0.546m (21.5”))

Stem (bow): is datum (zero) after the hull is turned upright
Foot bulkhead: 1745mm from datum (May be varied to suit leg length)
Rear of masik from datum: 2215mm (2362 mm)
Rear cockpit bulkhead from datum: 3065mm (3201mm)
Rear cockpit bulkhead to day compartment bulkhead 310mm (May be varied)
Rear cockpit bulkhead to centre of seat base 250mm (find this point by crossing diagonals from the corners of the seat base)
Internal cockpit 833 x 437mm (May be varied)
Height of rear coaming above keel 207mm
Masik knee clearance 280mm (May be varied)
Bow to foredeck buckle approximately 900mm
Stern to front of skeg box 920 mm (1015mm)
For those using a CNC cutting machine: the CAD layers in the plans are: 0 for CNC cut, 1 for centre line and gunwale line, 2 for all writing except identification numbers, which are on layer 3.

Fitting the sheer clamps

The sheer clamps are the light softwood battens, about 25 x 19 mm in section, which fit inside the gunwales to enable the decks to be fixed down. The sheer clamps and the masik (the curved foredeck beam) are now fitted as soon as possible, while ensuring that the maximum beam remains at 546 mm. Speed is again desirable to prevent distortion of the hull. If there will be a delay in fitting the sheer clamps and masik, some or all of the building forms can be temporarily placed inside the hull.
Sourcing wood of sufficient length to avoid any joints makes it much easier to produce fair gunwales. With a little persistence we have now found local sources of long lengths in both the U.K and the USA. If you find it difficult to source battens that long, two lengths can be scarfed together.
If necessary, kerf saw half way through the underside of the sheer clamps at bow and stern to enable the upturns to be completed without undue stress. (Kerf sawing is the technique of assisting timber to bend by sawing part way through it.)
Now dry fit the sheer clamps with many clamps (these can be bought in bulk very cheaply from EBay or Home Depot). Saw mitres in the sheer clamps at the stem and stern so they fit together:



The sheer clamps are fitted with the 25mm edge against the hull. Again clamp a batten across the gunwales to fix the maximum beam at 546mm (21.5”). The sheer clamps run flush to the top of the hull when aft of the cockpit, but from there forward the sheer clamps are allowed to rise up above the hull by up to 6mm. The 900mm nearest the bow can again be left flush to the top of the hull, as the foredeck becomes flat from that point to the bow. The protruding parts of the sheer clamps are later bevelled to accept the curved edge of the deck. Glue the sheer clamps in place with slightly thickened epoxy.



Epoxy filleting the seams.

Place neat epoxy fillets around the bulkheads, and up the inside of the stem and stern.
Epoxy fillets can be a ghastly mess or an elegant source of pride. We use wood mixing sticks (tongue depressors) to shape the epoxy, thickened to peanut butter consistency with West 406 colloidal silica. For narrow fillets, such as those around bulkheads, the stick is held vertically. For wider fillets, the stick is angled, or a piece of plastic sheet can be curved to serve as a shaper. We always pre-wet with unthickened resin the plywood surfaces where the fillets will be placed. Some of the fillets may then be fibreglass taped.

Taping the filleted seams

We tape around the foot bulkhead, but not those around the other two bulkheads, as they are comparatively lightly stressed. We use 50mm wide woven glass fibre 175 g/sq. metre tape.
The fresh fillets are very soft and vulnerable. If the glass tape is immediately placed on the wet fillets it is all too easy to dent the fillets with the fingers or brush. Once the fillets are dented under the tape we've been quite unable to flatten the surface. One option is to be very, very careful. Another option that we recommend is to leave the fillets for a couple of hours until they are less vulnerable, but before the dreaded "amine blush" coats the surface of the epoxy. Then wet the fillets with resin and lay in the glass tape.
Another method of causing fewer disturbances to soft fillets is to pre-soak glass tape with resin on the work bench before applying it to the epoxy fillets.
When cutting glass tape, make a half inch cut into the centre of each end to prevent unravelling
When wetting out glass cloth and tape, only use enough resin to make the tape clear. More will not be stronger. Epoxy resin is heavy, and it only gains full strength when glass fibres are added.

Making the masik (the curved deck beam)



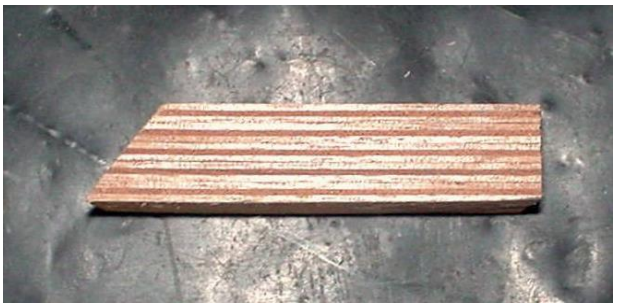
The masik is the main support for the foredeck. It also sets the clearance height for the tops of your legs when paddling. Some paddlers, especially rolling enthusiasts, like a low masik. Many others would prefer a high masik. This enables the knees to be lifted, and the strain taken off tight hamstrings, the cause of many a backache when paddling. The prototype Vember's clearance **under** the masik is 280mm (11")
The height of the masik can be altered by adjusting its curve, as described later. Increasing its height beyond 315mm (12.5") risks splitting the plywood deck when it is bent round the masik. The profile of the masik is drawn in the full size and CNC plans for 11" clearance. The A4 print PDF for the masik shows a full-size half masik with 11 inches clearance, ready to be turned over to draw the other half. If a clearance other than 11" is selected, then the top of the foot bulkhead may need to be adjusted to fit, as described in the Builders' Tips appendix.

To construct the laminated masik, cut 5 strips of 3mm ply from your offcuts, at least 700mm long and 45mm wide, to finish down later to 40mm.
Cover a section of your workbench with thick polythene to avoid the epoxy sticking the masik to your workbench.
Cut out the full-size profile of the masik from the plans and staple or pin it to your workbench.
Screw 5 blocks of approximately 50mm thick timber to the bench, three inside the curved line, and the end two outside. Clamp the five strips temporarily over the middle three to establish the correct position of the outer two. The line from the plans shows the outside of the masik.

Now withdraw the strips, brush the inner sides of the strips with slightly thickened epoxy, replace the strips in the blocks, and clamp lightly so that glue oozes very slightly all along the masik lamination joints. We eventually used about nine clamps, but only the first three are shown here:

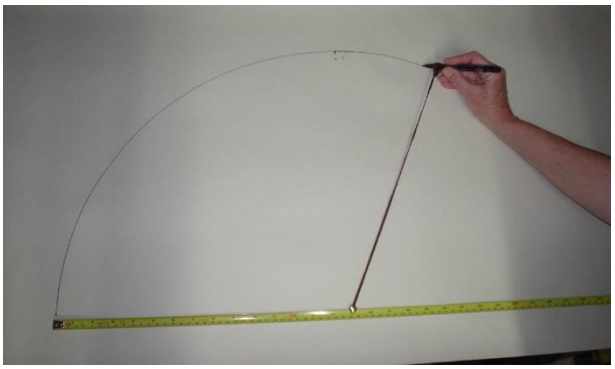


Leave the masik to set hard, preferably for five days at room temperature, before removing it from the jig, planing the edges to a finished width of 40mm. The edges of the masik will look like this offcut:
If the masik is released after only two days, expect it to spring back about 6 to 8mm in height. There is negligible spring-back after five days at 20 degrees C. (70 F.)



Creating a custom masik to change the knee height clearance

To form a custom masik height, if the standard heights are not ideal:

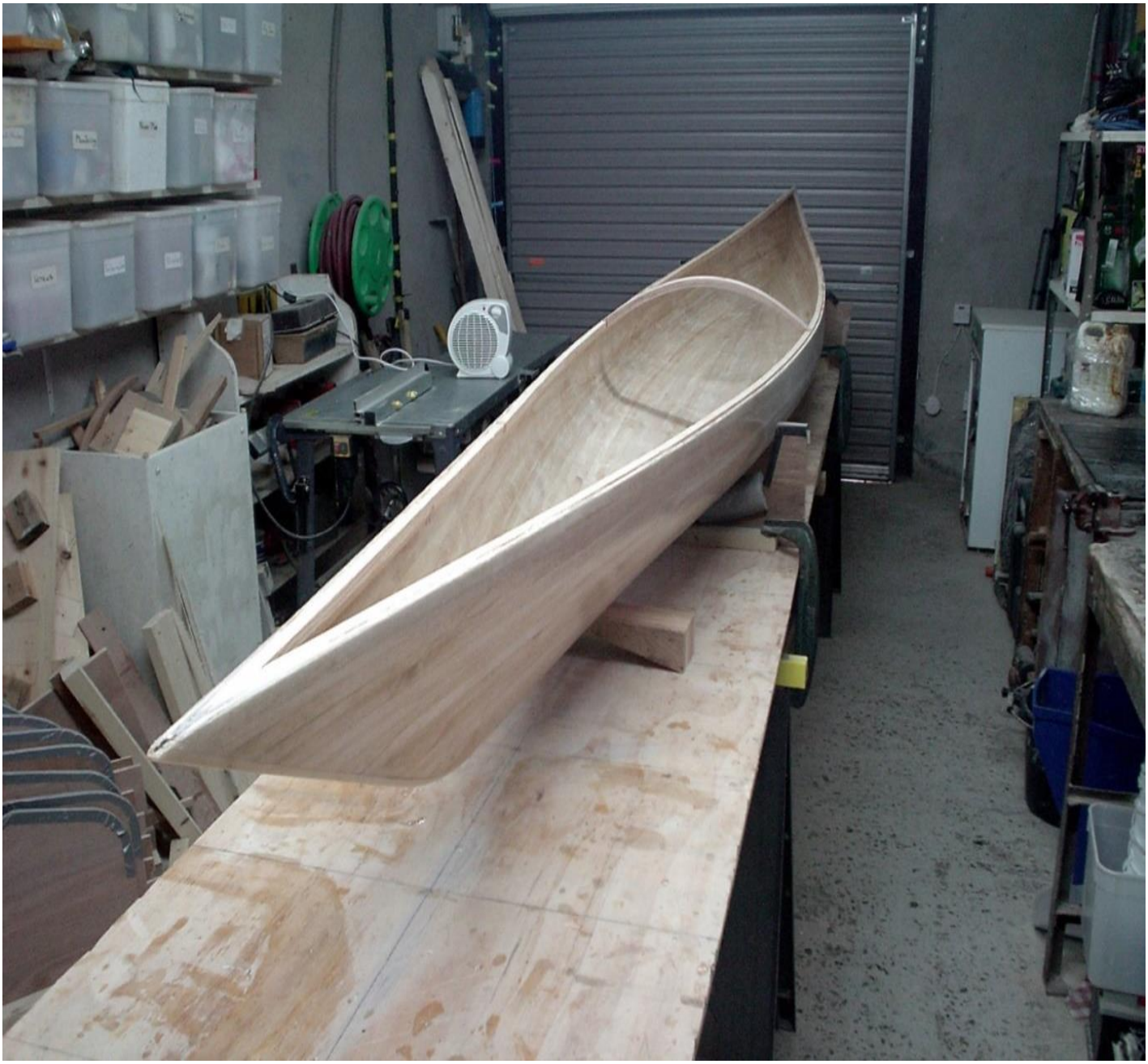


The curve is part of a regular circle, to place even stress on the curved plywood.
The radius of curvature of your desired masik, R, with height above the sheer line, D, and beam of the kayak at the masik, L (less 12mm, the two hull thicknesses), can be calculated from the equation: R = ((L^2) + (4*(D^2))) / (8*D)
The symbol ^ signifies “raise to the power of”, so L^2 = L*L = L times L.
Now tie a non-stretch thread round a drawing pin on some paper, and at a distance of R, roll the thread round a pencil. Draw an arc of a circle of radius R. For the prototype Vember we used L=534 (beam minus twice hull skin thickness) and D = 95 and hence R= 422 mm of string.

That is the **outside** profile of your modified masik. Now draw a chord (i.e. a straight line) of length L (beam at the masik) across the drawn arc.
Now proceed as for the construction of the standard masik.

Fitting the masik to the hull

The sheer clamps are notched to receive the masik, with its aft edge at 2215 mm from the bow. The masik is installed square to the centre line, of course, but it is not vertical. Its top surface is fitted parallel to the sheer line at that point, by sighting across the masik at the far gunwale. Saw and chisel a simple 45 degree notch, 40mm wide, out of the sheer clamps. Saw the two ends of the masik so that they fit into the sheer clamps, just butting up to the **inside** edge of the hull strips.
Glue the masik in position with temporary support, if necessary, from, for example, a couple of thin screws vertically through the sheer clamps. (Slightly rotate these screws every hour until the epoxy has started to set so as to avoid them being fixed in position.)



Fitting the bulkheads

Cut out the bulkheads to the plan profiles from the 3mm plywood using a fine-bladed jigsaw and a plane, and notch the top corners to fit round the sheer clamps. Every hull will be slightly different in shape, so testing the shapes before final trimming can be useful.

If no forehatch is to be fitted, cut a hole for a watertight access hatch in the centre of the foot bulkhead, to assist in drying out the forward compartment in case of leaks. Dry fit the bulkheads vertically and square at their positions and then superglue them in place with a few spaced drops of the glue.

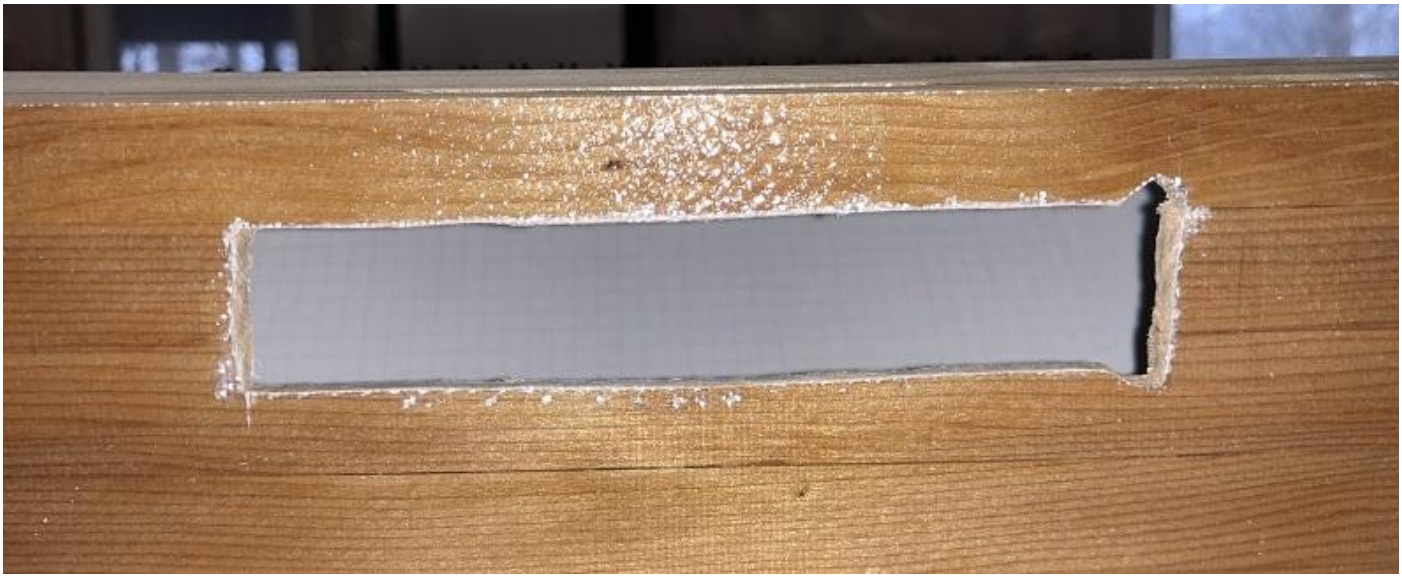
Rear cockpit bulkhead is at 3065mm (3201 mm) from datum (the bow). (May not be varied)

Rear cockpit bulkhead to day compartment bulkhead is 310 mm (may be varied)

Foot bulkhead at 1745mm (1881mm) from datum. (Do not fit this bulkhead at this stage if you are changing the foredeck height by varying the curve of the masik, or adjusting its position to suit your leg length.) The top of this bulkhead should be thickened with two extra strips of curved plywood to make a wider gluing area when the foredeck is fitted. If your feet will press against foam on this bulkhead, rather than foot pedals, consider using two layers of plywood and covering the bulkhead with glass cloth on both sides.

Preparation for fitting the skeg system

In preparation for fitting the wire operated skeg, drill a 6mm diameter hole in the top port corner of the aft cockpit bulkhead and of the day hatch bulkhead, ready to take the outer plastic tubing. Cut the opening for the skeg slider box in the port side of the hull where it would fall naturally to hand. When preparing to install the stern deck, coat the inside of these two holes and the skeg slider opening with epoxy resin.



Here the Kari-Tek slider box is fitted with the adhesive of your choice (we use Lexel or a silicon adhesive):



The skeg blade is formed from two layers of 3 mm plywood glued together while weighted down flat, and then cut to the plan profile so that the 6mm bolt just slides into the slot without excessive clearance. The skeg is sanded smooth. The actuating wire is epoxied in place, as shown in the plan. The epoxy must not touch the wire above the blade, as the wire must remain flexible:



The skeg box is cut to the plan profile from 3mm plywood, with 12mm square cedar internal framing. We now extend the side framing pieces above the box at least 100 mm above the top of the box. These are later notched into deck beams and trimmed to height. This braces the skeg box while maintaining maximum storage space in the stern compartment. The skeg pivot is a 6mm stainless steel rod or a bolt with the threaded section cut off, epoxied and glass taped into position on each side of the skeg box. If it is available, one should choose 316 grade stainless steel for superior corrosion resistance. The inside of the box is given two coats of resin, sanded smooth, before assembly.
The outside of the box is then glass sheathed after any sharp corners are rounded off. (At this stage one can fill the box with water to check the joints). The wire terminal brass compression fitting is then fitted into the top of the skeg, **ensuring that it is upright and square to avoid friction between the wire and the compression fitting.**



Optionally, one can glue a 1 or 2 mm thick nylon washer of about 40 mm diameter, slotted to take the pivot bolt, on each side of the skeg:



Now is a good time to test the operation of the skeg system on the bench. Thread the wire from inside the box and rotate the skeg into position. Pull on the wire with a gloved hand and check that the skeg lowers and rises smoothly. Adjust the skeg as necessary to achieve this.
Mark the position of the front of the skeg box on the keel line at 920mm (1015mm) from the stern, and drill a 2mm hole down through the keel.
Turn the kayak over and carefully mark out and jigsaw a slot aft of this 2mm hole so that the skeg box is a sliding fit into it. Turn the kayak upright, and fit the box into the slot so that its base is flush with the outside of the keel and plumb vertical, checked with a level across the gunwales showing horizontal. Epoxy fillet and glass tape the box inside the hull, ensuring it stays vertical and at the correct height with temporary bracing as required. The skeg box must be able to cope with the stress from the skeg colliding sideways with rocks or a beach. To help avoid this stress causing damage to the surrounding thin hull panels, the extended vertical framing is notched and glued into two deck beams when the epoxy has set:



The next day, install the plastic outer tube into the fitting on top of the skeg box so that it just reaches the inside of the box. Slightly tighten the nut. Thread the plastic tube through the holes in the bulkheads and cut it to length so that it just reaches into the end of the slider push-fit connection while allowing a fair and easy curve in the section by the skeg box. The tube can be fitted to the inside of the sheer clamps using plastic P-clips and stainless steel screws.



Now remove the plastic tube from the fitting on top of the skeg box. Thread the end of the skeg wire up through the skeg box and into the end of the disconnected plastic tube. Thread the wire all the way through to the slide box and through the tube until it emerges from the end of the tube outside the hull. Reconnect the plastic tube to the fitting on top of the skeg box, and carefully tighten the nut.
Pull the wire so that the skeg is fully retracted. Push the actuating button forward until there is just enough space to place a finger in front of the button. Tighten the grub screw with a 2.5mm Allen key (hex wrench in the USA) to fix the wire in position. Do not cut the wire to length at this stage. Make a final check that the skeg system is operating smoothly by laying an off cut of plywood across the top of the deck beams by the skeg box, and weight this to simulate the restricted space left for the plastic tube once the stern deck is in position:



Now remove the skeg and wire so that shavings and sawdust from later trimming the protruding base of the skeg box do not jam the system.

Stern deck beams

The other stern deck beams are installed in positions to suit the hatches you will fit. The rear of the stern hatch opening should be level with the front of the skeg box, to allow access for maintenance, and for fitting the operating cable. Fatten up the aft cockpit and day compartment bulkhead tops on the day compartment sides with offcuts from the sheer clamps to give a larger land for gluing down the stern deck. We also added two fore-and-aft short beams in the day compartment area to suit the day hatch, and to strengthen this area. All the beams are notched at 45 degrees into the sheer clamps or beams, and epoxy glued into position.





Glassing inside the cockpit

One option now is to cover the floor of the cockpit with an extra layer of glass cloth. This is easier to do before the decks are fitted. An extra layer of glass adds to the impact resistance of this vulnerable area and protects the floor from grit and sand on the paddler's feet, but there would be a weight penalty. Delineate the area you wish to glass with masking tape, pre-wet the hull with the minimum resin, lay in the cloth, (pre-cut to shape), and stipple a minimum of resin into the cloth to make it go clear. Trim the edges of the cloth along the edge of the masking tape with a utility knife when the resin is firm but not hard.

Fitting the decks

The 3mm plywood decking is in three parts:
1. The foredeck, which reaches from the bow to the centre of the masik.
2. The stern deck, which incorporates the cockpit, and extends aft from the centre of the masik for the full length of the plywood sheet.
(3. A small filler piece to complete the deck to the stern if a Vembex is under construction.)

The plans do not show the outline of the decks, as the plywood sheet can be laid over the part-built kayak, and the desired shapes transferred directly. Alternatively, use sheets of newspaper, taped together, and take a pattern from the deck area. Allow 20mm overlap on the gunwales. Cutting the panels too small would be bad news.

Material for fitting the decks

Materials for the decks:
One sheet of 3mm (1/8 inch) quality waterproof plywood (BSS 1088 marine is best) available from CLC.
Elite marine plywood from Robbins of Bristol, UK or Fyneboatkits is perfect.
MAS low viscosity epoxy resin and slow hardener.
Disposable gloves - box of 100
Many disposable cups for mixing, e.g. used yoghurt pots
Disposable cheap 25mm (1") paint brushes.
Tongue depressor mixing sticks for shaping epoxy fillets.

Thick plastic sheet is useful for preventing epoxy adhering to workbench, workmate, etc.
Hand wipes (We've used Ever-Build Multi-use Wonder Wipes in the UK)
Rags and paper towel roll
Polyurethane varnish to protect the epoxy from ultra-violet light
Optional: Super-glue with accelerator. Evo-Stik Mitre adhesive and E-Z bond have each worked for us.

Tools for fitting the decks



Before fitting the decks, it is helpful, but not essential, to make some 3mm plywood tools. From left to right in the photo:

1. The far left tool is for measuring the varying angle at which to bevel the sheer clamps. The cut-out by the wing nut accepts the overhanging part of the deck when it is dry fitted.
2. (The next tool is not required for Vembers. It is for Shrikes, if the deck/hull joint is taped.)
3. The next right transfers the position of the gunwale onto the deck to facilitate marking the deck for trimming to size by panel saw, and then block plane.
4. The tool at far right has the same notch to clear the deck, and a beak to extend the point beyond the gunwale edge to the mid-point of the sheer clamps. This is to mark the position of any screws for attaching some deck line fittings.

Bevelling the sheer clamps

The decks are glued down onto the top of the sheer clamps, which must be planed to the correct angle. This angle varies throughout the length of the sheer clamps. Use a block plane and/or a spokeshave.
From the stern, along the stern deck and up to the rear of the cockpit, the sheer clamps are planed horizontal, so that the flat stern deck sits straight on it, and this should be done at this stage to facilitate the accurate positioning of the stern deck beams.
From there, forwards, the angle varies to accommodate the landing angle of the curved deck plywood. The 900 mm of the sheer clamps from the bow aft can remain level with the gunwales. The angles at various points can be found by temporarily fitting the plywood foredeck, and measuring the angles at various points with your sliding bevel, which you can make from plywood, as previously described. In traditional boat-building the angles at each point were then drawn on the work table at each point, or scribed onto a "bevel board", a plank of wood used as a form of notepad. Check frequently that the plywood lays symmetrically across the masik. The loose clamps at the ends should allow the edge of the plywood to move slightly across the masik so that the centre lines of the masik and plywood coincide. When the plywood reaches the gunwales, check that the plywood sits comfortably on the previously angled sheer clamps. If necessary, remove the plywood and adjust the bevel of the sheer clamps.
The masik can also receive a light bevel fore and aft to facilitate a neat joint between the fore and aft decks on the centre of the masik. Draw a pencil centre line across the masik from gunwale to gunwale. A straight edge between the centre line of the masik and the aft cockpit bulkhead will guide the maximum amount that can be bevelled, while not touching the centre line. Similarly, a straight edge from the centre line of the masik to a point about 900 mm from the bow (where the foredeck buckles) will give a guide to the amount that can be removed from the front half of the masik. Do not touch the pencil centre line on the masik.

Fitting the foredeck

The foredeck is fitted before the stern deck. The plywood sheet extends from the centre of the masik to the bow of the kayak. Draw a line along the middle of the masik from gunwale to gunwale. Draw a line across the mid- point of the masik.

Lay the plywood sheet over the foredeck, with one short edge lying along the centre of the masik, across the kayak.

Tightly clamp the plywood at the centre of the masik in order to locate the plywood while the fitting process continues. Lightly mark the edge of the plywood at the centre mark on the masik. Use a scrap of plywood under any clamps to avoid damage to the plywood.

Bend the plywood across the masik, and clamp the plywood loosely to the ends of the masik. Mark the edge of the plywood at the centre mark of the masik. Use wooden beams under the ends of the straps to avoid snapping off the edges of the plywood. Bend the plywood towards the gunwales and mark a trimming line underneath the plywood so as to leave a safe margin, about 20mm, hanging over the gunwales. Remove the plywood, cut to this line, and then replace the plywood and the clamps.



The plywood should buckle at a point about 900mm from the bow, depending on the height of your masik. The plywood will dictate the position.

While doing this, check that the top of the foot bulkhead is not obstructing the bending process. Trim the top of the bulkhead if it is distorting the deck. Adjust the height of the foot bulkhead as necessary if the masik height has been changed. Before the plywood reaches the gunwales, check the size of the excess overhanging plywood. If it is more than about 10mm, it is worth trimming it a little closer, to avoid the excess plywood snapping across the gunwale as pressure is applied, particularly at the corners by the masik. Now strap a beam across the plywood just forward of the buckle, and check and adjust your sheer clamp bevels:





Refit the plywood with its straps and beams, and check that it is resting closely on the masik across its entire length. Fit several clamps across the masik. To show the gluing area, draw a line across the masik to mark the edge of the plywood, and mark underneath the overhanging edges of the plywood. Slowly and evenly release the straps. At this stage, fit any under-deck fittings such as Maroske fittings (Google if you're not familiar with these) or our 3D printed fittings (as shown in the photo below and later described in the Builders' Tips Appendix) that will be inaccessible once the foredeck is installed:



If you have decided to fit a hatch in the flat forward section of the foredeck, now is the time to install any necessary deck beams. Apply masking tape to the hull below the gunwale to protect the hull from any excess epoxy. Apply thickened epoxy to the tops of the sheer clamps and the foot bulkhead and part of the masik. On the bench, roll a coat of epoxy resin onto the underside of the deck panel. The panel is immediately fitted, repeating the clamping and strap tightening until glue oozes all along the deck/hull joint. Scrape off the excess epoxy while it is partly set, **particularly along the top of the masik where you will later need to make an accurate joint between the fore and aft decks**.
Get creative about applying weights to hold down the flat areas. We have used a trolley jack with pots of paint, and here we use weights for weight-lifting:



When the glue has set hard, the straps are removed, and the edges of the plywood are trimmed and sanded back to the gunwales, while protecting the hull with masking tape:



Turn the kayak upside down, and fillet and glass tape the joint between the top of the foot bulkhead and the underside of the foredeck. (The optional hatch is fitted earlier if no foredeck hatch is fitted.)



This is also a good time to plane and epoxy any protruding parts of the skeg box flush with the outside of the keel, after using a perimeter of masking tape to warn when the plane blade gets too close to the glass cloth of the hull, and temporarily blocking the box opening to prevent shavings from dropping into the box. If you wish to install a knee tube under the foredeck, now is the easiest time to fit it. Details of the knee tube construction and fitting are in the Builders' Tips Appendix.



Completing the installation of the skeg

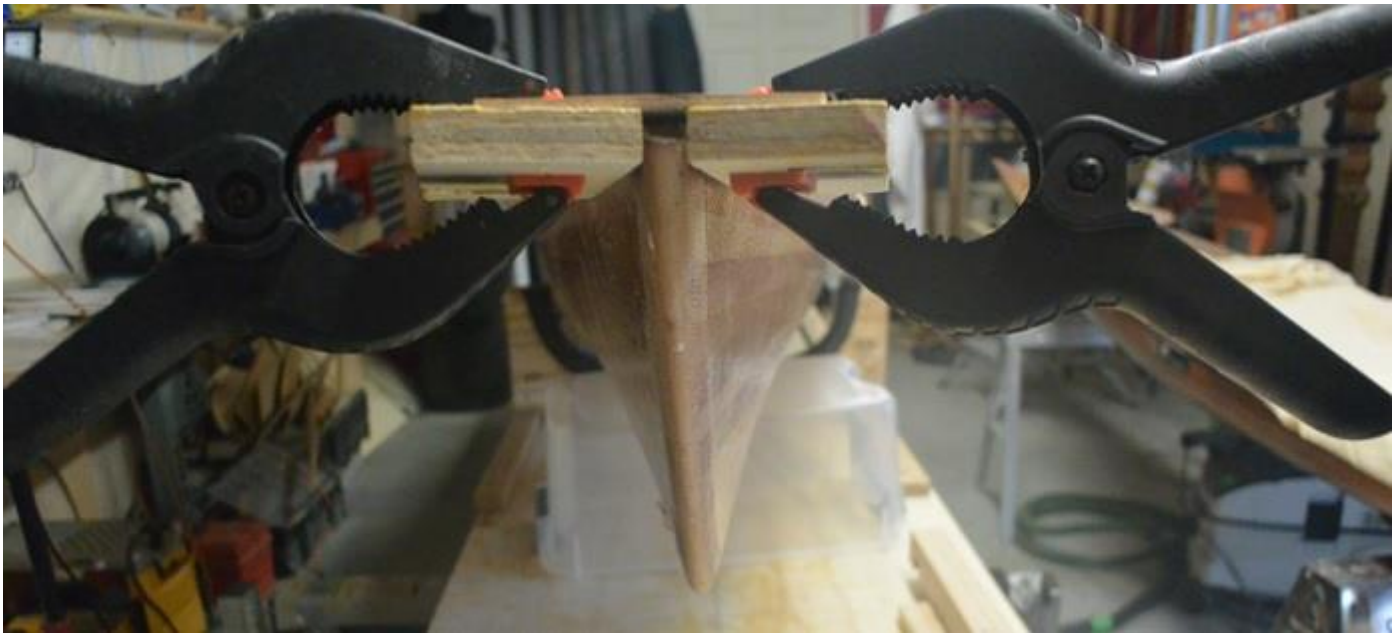
When the epoxy has set hard, turn the kayak over. The skeg wire can now be wiped with a cloth wetted with silicon lubricant spray. Keep the lubricant away from the section where the wire will be cut and from anywhere that will be epoxied or varnished. The skeg and wire can now be fitted and the wire cut to length after smearing with quick-setting epoxy or CA glue the section to be cut through when the epoxy has set. Pull the wire so that the skeg is fully retracted. Push the actuating button forward until there is just enough space to place a finger in front of the button. Tighten the grub screw with a 2.5mm Allen key (hex wrench in the USA) to fix the wire in position. Check that the skeg operates correctly. Cut the wire level with the front of the actuating knob.

Fitting the stern deck

The stern deck main panel uses the full diagonal length of the plywood sheet, and stretches from the masik, aft to a short distance beyond the stern. (Alternatively, a short panel and a butt strap can be added to make up the length). Cut a panel to fit the sterndeck, with a 20mm overlap on the gunwales, as for the foredeck. Cutting a smaller approximation to the cockpit opening in the centre of this panel will ease the bending strain required, so, from the plans, scribe a shape at least 50mm smaller than the final opening, and cut it out on the bench. (At this stage it is useful to make a rough tracing of the stern deck, using newspaper or wallpaper, to outline the sheer clamps and all the stern deck beams. This can be a great help in remembering later where to cut the holes for the hatches.) Mark the centre of the short side of this panel, and preserve the previous marks on the centre of the masik and on the foredeck plywood. Ensure these marks coincide throughout the remaining procedure. Place weights on the length of the stern deck aft of the cockpit, to mate it flat to the sheer clamps. At the aft cockpit bulkhead, strap a beam across the kayak so that aft of this bulkhead the deck is flat. Place the other four straps round the hull, and gradually tighten them. The strain on the plywood overlaps is particularly great at a point just forward of the strapped beam and at the masik, so trim the overlaps to about 10mm at these points.



Note that the two panels do not meet all along the masik. Using a pair of dividers or compasses, scribe the shape of the foredeck edge onto the edge of the stern deck panel. (Now you are pleased that you carefully cleaned this area of epoxy when fitting the foredeck.)
Check the angles of the bevelling and note where adjustment is needed to fairly land the panel onto the sheer clamps.
The aft end of the panel tends to move during the fitting process. Locate it by clamping a small wooden block under the overhanging excess plywood each side of the stern.



Remove the stern deck panel, and plane the marked edge that was on the masik. Refit the stern deck panel, and gradually tighten the straps until the panel touches the sheer clamps along its whole length from the masik to the cross-strapped beam.
Repeat the removal and trimming procedure until you are happy with the fit. Place any under-deck fittings at this stage. Optionally, one can now place a temporary screw down through the deck panel into the centre of the masik to assist in locating the deck panel before gluing. This screw can be positioned discreetly under where the cockpit rim will later obscure the screw hole. Slightly rotate the screw every hour until the epoxy is partly set, in order to avoid trapping the screw permanently.
Remove the panel, immediately roll a sealing coat of epoxy on its underside, apply thickened epoxy glue to the tops of all the stern deck beams and the sheer clamps, and glue the panel into position, using weights and straps to produce a bead of glue all around the joint.



When the glue has set, the stern deck panels can be trimmed to the gunwales.
Put a light radius on the edges of the gunwales with a block plane and 80 grit sandpaper. If you wish to cover the decks with fibreglass cloth, now apply glass cloth and epoxy over the entire deck and down the hull for 25 mm. This is much easier at this stage, rather than after the cockpit is constructed. However, with such a wide gluing surface on the sheer clamps we now consider that this joint requires no reinforcing. Damian form Plymouth, UK, decided to cover the decks of his Vembex with glass cloth:



Cutting the cockpit opening and building the cockpit coaming and leg braces.

Cockpit opening cut in deck…….833 mm, later reduced by two 3mm upstands to 827 mm
Cockpit internally…… 827mm long x 437mm wide (32.5” x 17.2”) (These dimensions can be varied to suit.)
Clearance height, keel to underside of masik at front of cockpit…… 280mm (11”) (This can be varied.)
Keel to top of rear rim of cockpit with standard gunwale height… 207mm (8.2”)
The internal width of the cockpit can be adjusted to suit the physique of the paddler. Try other cockpits to see which width will suit you. The internal length of the cockpit can be decreased if desired, perhaps to Ocean Cockpit dimensions. (See Builders’ Tips appendix for construction details for an Ocean Cockpit)
The size, shape, and position of the leg braces in the cockpit are vitally important for the secure handling of any sea kayak. The leg braces shown on the Vember plans suit many paddlers, but they may not suit you. We suggest you try out other kayaks by sitting in the cockpit, supported on soft cushioning, in order to design your personalized leg braces. See the Builders Tips section for further discussion of leg braces. We have now extended the braces on keyhole cockpit kayaks so that we can only just fit each leg in the gap between them when entering the cockpit when wearing a drysuit. The gap between the plywood braces on my Vember is now 140 mm, and only 115 mm between the shaped foam padding.
From the plans, or to your own preferences, draw the inner profile of the cockpit lip centrally on the deck, and cut accurately to this shape. We do this with a utility knife, but one can use a jigsaw with a downward cutting blade (sometimes known as a laminate cutting blade) in order to protect the top veneer of the plywood. The front edge of the cockpit opening is at the aft edge of the masik, and the aft end of the opening is against the cockpit aft bulkhead, making the opening length 833 mm. The cockpit final opening has a length of 827 mm when it is reduced by the 3 mm upstand at each end. Slight variation in these measurements can be compensated later by adjusting the width of the cockpit lip.
In the photo below we have removed the leg brace wings so that braces can be positioned further aft when the particularly small owner has sat in the cockpit.



The cockpit coaming upstand is now created by bending the 3mm plywood around the inside of the cockpit opening:
Cut 30mm wide strips of the plywood of sufficient combined length to form the perimeter of the opening. The long straighter sides of the upstand can be created from these strips, superglued (using spray accelerator, an excellent invention) into position. The upstand is set vertically, and flush to the underside of the deck plywood. At certain points the upstand will extend a little below the deck, and these areas will be filled later from behind the upstand.
At the sharp bends, the plywood is cut almost through from the outside at intervals of 10mm, using a fine-toothed saw and the cutting board that was used for the strips. It does not matter if the plywood breaks into shorter sections. In fact, this makes it easier to deal with the compound curves. Fit them all in position with super-glue and accelerator.



The plywood upstand serves merely as a former for the next stages, the strengthening of the upstand with epoxy and glass tape. Coat the outside of the upstand with resin, and place a neat fillet of thickened epoxy around the perimeter joint. Then epoxy a strip of 50mm glass tape all around the outside of the upstand across the upstand/deck fillet. Next, epoxy a length of glass tape around the vertical inside of the upstand. Snip the outside selvage edges of fibreglass tape on internal bends as at front inside of cockpit coaming. Trim the excess width of tape when the resin is part set. Now that the upstand is strengthened, the top edge can be gently planed down to your chosen height and to produce a slight concave curve when viewed from the side, with the eye at hull level. Without shaping, the top edge will be almost straight, an inelegant profile. The concavity need only be very slight, as shown here with a plastic batten to indicate the curve.



In the photo below the leg brace wings have been retained in the deck plywood:



The 6mm thick coaming top lip is now formed from either two layers of 3mm plywood or from 6mm plywood, if available. The top lip should be cut out with the profile shown in the plans, but with the leg brace bulges removed, as these are solely for drawing the final cockpit hole in the deck. Allow at least 10mm extra on the inside, to allow for variations in construction of the 3mm upstand.
Lay the rough cut lip in position on the upstand, draw round underneath and inside, and now cut just inside this line. Glue and clamp the lip until glue oozes slightly all around the joint. (Alternatively, CA glue the lip onto the rim. This avoids a lot of clamping and weighting)



Clean off the excess glue before it has set. When this has set, turn the kayak upside down, and fillet and glass tape the joint between the upstand and the lip. When this has set, turn the kayak upright. Round the outside edge of the lip with sandpaper. The inner edge of the lip can be profiled with a router, or with 80 grit super-glued to a length of 100 mm diameter drain pipe:



The top of the cockpit lip and the inside of the upstand can now be coated in resin and glass cloth for abrasion resistance and strength. As this involves double curvature, the woven glass cloth should be bias-cut (45 degrees to the strands), using a rotary cutter. Alternatively one can use twill woven cloth.
Bias cut glass cloth should initially be cut well oversize, as it is dimensionally unstable. (Use masking tape where the cloth is to be cut, and use a craft knife or rotary cutter to slice through the centre of the tape)
We are currently building a Vember for a slim lady family member, so we have fitted a particularly narrow cockpit with internal width 360 mm maximum. This leaves a wide area of unsupported side deck, just where one is liable to lift the kayak when ashore and where a strong knee lift in a roll would be applied. We therefore fitted a pair of plywood braces to connect the cockpit rim to the hull:



These braces, which are first glassed on both sides, need to be strongly attached to the cockpit rim and to the hull. Four layers of glass cloth in diminishing sizes cover the areas of the hull where the braces attach to the hull, starting at 130 mm diameter and reducing to 120, 110 and 100 mm, with the smallest on top. The top of each brace is secured by two layers of glass cloth stretching from the top of the cockpit upstand down to and across the hull reinforcing glass. Tape the back of the joint between the hull and the brace. When this has set, turn the kayak upside down and tape the joint between the deck and the top of the back of the brace. This seems a lot of fiddly work, but these are the only joints we have seen fail on any of our designs, and on each of these three occasions the joints had not been taped, but had relied on epoxy fillets, which are strong in compression, but comparatively weak in tension.

Strengthening the leg braces

The protruding wings of the leg braces can be highly stressed, both when paddling and when lifting the kayak. They can be strengthened as follows: Epoxy a 3mm plywood plate, with the same shape as the wings, under each wing, extending under the side decks to the sheer clamps, so as to strengthen the wings. Further strengthen the wings by covering both sides with glass cloth, and add a large epoxy fillet between the top of the wings and the cockpit rim upstand. Someone will eventually pick up the kayak by just holding a wing, so they need to be strong enough to withstand that abuse. Use contact adhesive to secure shaped hard Minicell foam under each wing to create a hook effect for the inner legs.



Alistair from Cornwall cut a radial pattern part way through the wings, bent them down so that they fitted his legs, and used CA glue to preserve the shape, before coating with glass cloth on both sides:



Cutting hatch openings

The prototype Vember in the photographs was designed as a light-duty day cruiser, so no fore hatch is fitted, and the forward compartment is filled with an inflated buoyancy bag. There is a hatch in the foot bulkhead. A foredeck hatch could be fitted well forward, on the flat section of deck. For day trips, the day compartment provides adequate stowage, and no fore hatch is necessary. However, experience on short camping trips shows that the hatch in the foot bulkhead is suitable for stowage of items such as tents, and this has the advantage that a foredeck without hatches is leak-proof, as well as being elegant and strong. If you can't reach an item, then, with the kayak on the beach, lift the bow and the item drops towards the hatch. Sometimes the simplest solution is the best! The stern compartment hatch is required for access for maintenance if the optional skeg is fitted. The day hatch is the small disc immediately behind the cockpit. A day compartment is very useful for keeping small items together, especially on day trips, when these items are all that are carried.

The hatch openings are now cut into the stern deck. We cut a small hole in the approximate centre of each intended hatch opening, and, with a short steel rule, established the position of the deck beams, and hence the line to jigsaw the accurate holes for the hatches. We used the inside of the hatch rims as templates to draw the lines.

As the stern deck immediately aft of the cockpit can be heavily loaded if a paddler sits on this area, we glued a short extra transverse beam just forward of the day hatch opening, clamping temporarily through the opening.

Finish coating the hull and deck



Sand any rough edges from the deck, and turn the kayak over to begin finishing the hull. You must now choose how you wish to finish the surface of the kayak. In the earlier photos with the hull constructed from the pale Pawlonia, Nick just sanded the whole kayak, rolled and tipped a coat of epoxy over the complete hull and deck, with no varnish or other UV protection. He justifies this on the basis of the lack of sunshine in the U.K. and the fact that the kayak will be stored in a shed with no windows. Also, any scratches or other damage are simply finished with epoxy resin, with no concern for layers of varnish. If only applying epoxy resin without any covering varnish or paint, use firm foam rollers to coat the panels with an even, thin, run-free coat. Avoid the use of brushes where possible, to avoid runs. Immediately follow the rolling with tipping off with long strokes with a piece of the roller, held to stop it from rotating. This removes air bubbles. The resin finish should not be applied when the temperature is increasing, as bubbles can appear.

Any amine blush must be washed off the epoxy coating with soapy water, and the surface lightly sanded, before any gluing, glassing, varnishing or painting.
One must attempt to create a dust free environment for varnishing and painting. We use fly spray in the workshop the night before painting or varnishing, after vacuuming the floor. The cockpit floor may be painted with, for example, International Interdeck non-slip deck paint.
Christopher in the USA, on the other hand, rolls on the epoxy fill coat, and then applies seven coats of varnish, with much sanding in between, leaving a superb high gloss finish.



Here’s how to do it:

1. The hull will be finished first, so apply 50 mm wide masking tape to the edge of the deck so that resin and varnish runs from the upside down hull do not touch the decks.

2. Turn the kayak over and sand the hull using a random orbital sander and 120 grit sandpaper.

3. Clean the hull with de-natured alcohol and then with a tack cloth.

4. Apply varnish with a 50 mm (2 inch) foam brush. Brush horizontally and then vertically twice, and then finish off horizontally in 300 mm sections. Use a UV protecting spar varnish, such as Pettit Z-Spar Captain’s Varnish.

5. Allow the varnish to harden according to the manufacturer’s instructions, gently sand with 220 grit to remove high spots, clean as at 3, above.

6. Repeat a third coat using 220 then a fourth and fifth using 400 grit.

7. Determining which coat is the last is a personal choice. Once the last coat is set, use 1000 grit sandpaper followed by 00 grade bronze wool. Afterwards use rubbing compound with an electric polisher followed by polishing compound. (e.g. the 3M Perfectit products).Leave the varnish to harden for a couple of days, turn it over and repeat the process on the deck:



Outfitting

Secure the hatches into the deck and foot bulkhead with silicone adhesive and stainless steel bolts. In the U.K, SeaWorld and Barton hatches have proven to be 100% watertight, as claimed. The screw threads must be kept clean, and a smear of silicon grease on the sealing rings will help keep out any water. A foam seat can be purchased or one can be carved from a Minicell foam block, as can a backrest, as shown below. **The centre of the seat is 250 mm in front of the rear cockpit bulkhead**. Changing the seat position even by a small distance position can have drastic effects on the handling of such a light and manoeuvrable kayak. The seat should be as low and thin as possible in order to assist stability of the kayak. The plans include a form shape for the foam backrest we use.



Deck fittings

If under-deck fittings have not been fitted, deck fittings to hold perimeter lines can be made from 85mm lengths of 25mm wide polypropylene webbing tape, folded into loops. The deck loops can be fixed to the sheer clamps with 19mm (3/4") pan head M4 (no.8) stainless screws through 20mm M4 penny washers. Constructing glass and epoxy Maroske fittings is an interesting exercise. (An internet search will find details of these). However, we now use 3D printed fittings, glued under the decks after two holes are drilled in the plywood. The data file to drive a 3D printer is included in the plans download. Descriptions of all these options can be found in the Builders' Tips appendix.

Here some 3D deck fittings are seen as fitted to Vember:



And 3D printed deck fittings for the bow and stern lines:



And now she's complete:







So let’s go paddling!





Appendix 1: Vembex, the Vember Expedition version



In order to give greater carrying capacity and maximum speed than Vember, Vembex extends the length by 10% to 5.35m (17.53 feet), while maintaining the same beam. The construction follows the Vember Build Manual, with the below listed changes:

1. The transverse temporary forms are spaced at 330mm rather than 300mm.

2. The 110% stretched (only in the horizontal x-axis) bow and stern longitudinal temporary forms are taken from our download. The resultant form structure was assembled on our particular 4480mm workbench with the bow form extending 300mm beyond the bench at the bow, and 185 mm at the stern.

3. The (optional cove-and-bead) strips quantity is increased to 250 metres.

4. The aft edge of the masik is at 2362mm from the bow (datum). This is 839mm in front of the aft cockpit bulkhead to allow for the 833mm cockpit opening plus the 3mm plywood rim at each end.

5. The aft cockpit bulkhead is fitted at 3201 mm from the stem of the completed hull.

6. The front of the skeg box is 1130mm aft of the aft cockpit bulkhead. This is 1015mm from the stern.

7. The foot bulkhead position may be varied to suit, but we fitted it at 1320mm in front of the aft cockpit bulkhead, as with the standard Vember. (This position is also 1881mm from datum in Vembex).The height of this bulkhead may have to be adjusted to accommodate your chosen position and the height of your masik.

8. The stern form is 458 mm long x 311 mm high. The bow form is 931 mm long x 358 mm high.

9. If built to the same specification as the prototype Vember, the weight will increase to approximately 38 pounds. (17 kilos)



The below photo shows Vembex, the Vember Expedition option, with a total load of 112 kg (246 pounds), comprising an 83kg (182 pounds) paddler and a 29 kg (64 pounds) load of gear in the compartments.



Damian completes a hand roll:



Appendix 2: Adjusting the length and beam independently

As stated in the body of this Manual, the length of your kayak can be adjusted by changing the spacing of the temporary forms. The plans download includes the plans for the standard Vember with 300 mm spacing, and the stretched bow and stern longitudinal forms for Vembex, with 330 mm spacing. For other lengths the below table gives some suggested spacing:

Spacing	Length	
340	5508	
330	5346	(Vembex)
320	5184	
310	5022	
300	4860	(Vember)
290	4698	
280	4536	
270	4374	
260	4212	

In addition, the bow and stern forms must be stretched or shortened longitudinally by the chosen percentage. This has been done for you for Vember and Vembex in the downloads, but for other lengths you will need to change the x-axis dimension (along the keel) by the chosen percentage, while retaining the y-axis dimension (vertically). Some printers offer this facility. Alternatively, the A4/letter downloads include offset measurements for the bow and stern templates. The beam of the kayak can be adjusted by changing the percentage print of the paper plans of the temporary forms. Both Vember and Vembex use a beam of 546 mm (21.5 inches), but the below table gives some suggested scaling for other beam measurements:

Print %	Beam	
110	601	
108	590	
106	579	
104	568	
102	557	
100	546	Vember
98	535	
96	524	
94	513	
92	502	
90	491	

If you are using the continuous roll printing facility, this will alter the dimensions of the bow and stern forms, and you will wish to correct the length along the x-axis, as described above. If you are changing both the length and the beam then you will have to adjust for changes in the dimensions in both axes. The need for this will vary according to the facilities of your printer option. (An example of this procedure is a local experienced paddler who intends to build a Vembex with the beam reduced by 10%. The form spacing remains at 330mm, and the form plans are downloaded at 90% scale for the reduced beam. The bow and stern forms must then be corrected. The 10% reduction in freeboard can be reversed by adding an extra strip at the gunwale level during construction, if desired.)

Appendix 3: Builders' Tips

Ocean cockpit construction details

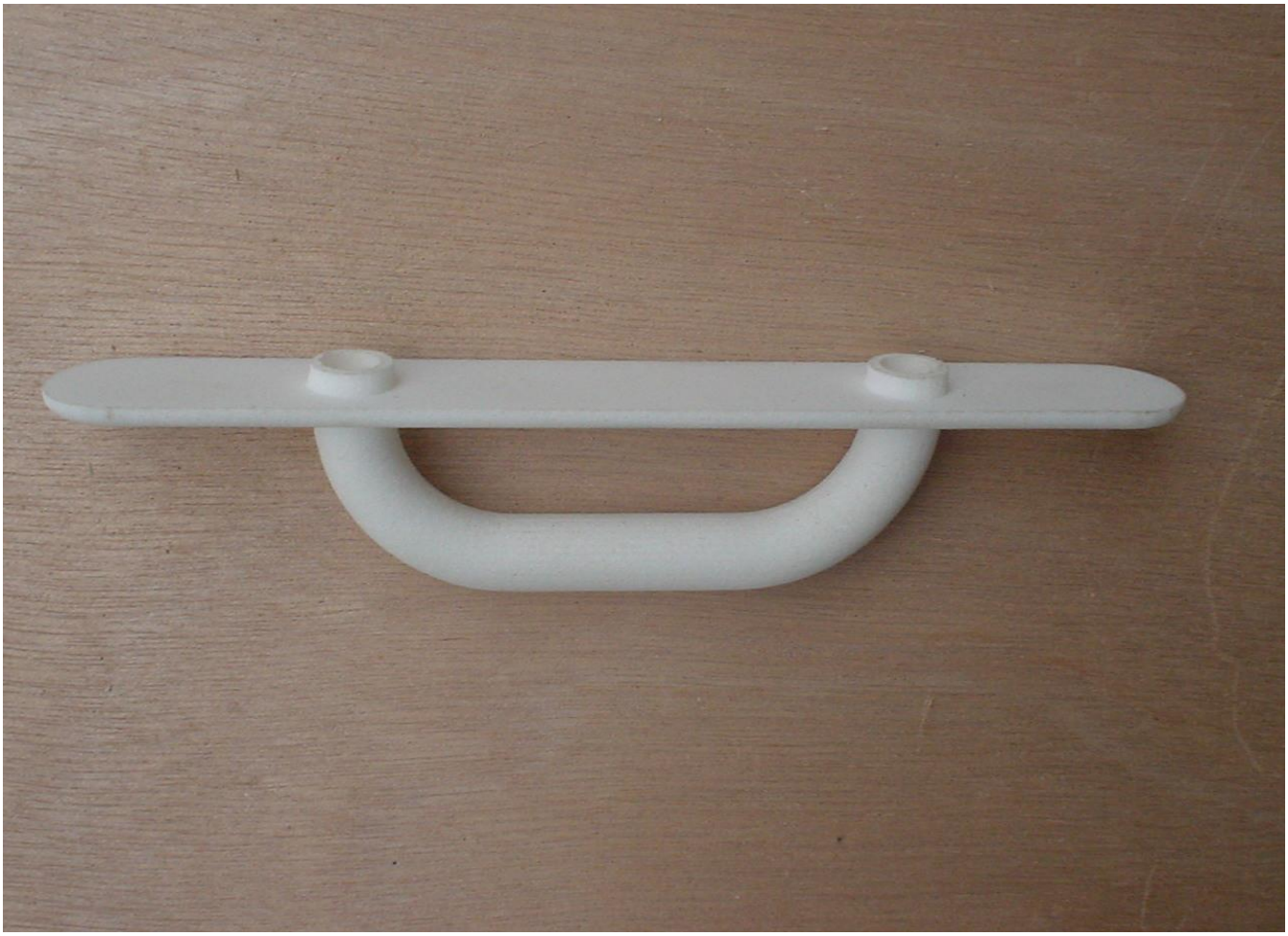
Christopher favours the small "ocean cockpits", and these necessitate some changes to the position and height of the masik. Here's how he calculates these changes:

"The only area that matters "upstairs" is where you end up sitting and where that places your knees. I focused first on identifying where I would be sitting. This is a fixed point from the measurements list, and it should not be changed. I placed the hull (just after fitting the sheer clamps) on the ground resting on cushions and very gently sat inside it. By doing this I was able to identify where I wanted the masik, or deck beam; an inch behind my knee caps. Unlike the first Shrike, I defined the position of the deck beam based upon my body rather than the size of the plywood. It would be possible to do it both ways by using two deck beams, one for the plywood joint and the other for the knee beam (masik) but I decided it would be simpler to slide the plywood aft to the position of the masik beam leaving a small triangle at the bow rather than attempt to construct and align two curved deck beams. The other critical measurements are the deck height and cockpit length, and they are related. You need to decide upon one and the other is derived from it. I chose a cockpit length of 19inches as this was similar to my other kayaks and allowed me to share spray skirts between them. I then used my body to decide how high the deck would to be to curve to allow my knees to get under the deck at the front of the cockpit while my butt was sitting on the back of the cockpit rim (which remember will be slightly raised above the deck). This sounds complicated but it isn't really. I sat on a plank straddling the stern deck and I rested another across where the masik was going. I was then able to adjust it and approximate how much clearance I needed above the gunwales (70mm)."

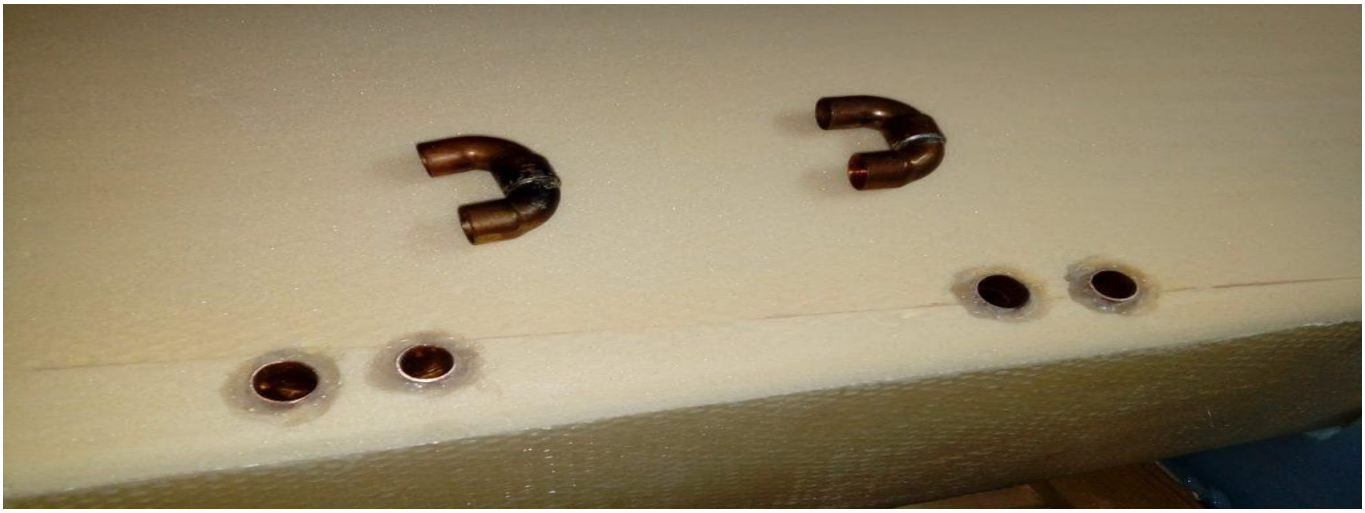
Deck line fittings

The standard low cost and simple solution deck line fitting is to use short lengths of folded tape screwed into the sheer clamps, as described earlier in the Build Manual. There are other options:

We use the 3D printed fittings whose printer files are included in the plans download:

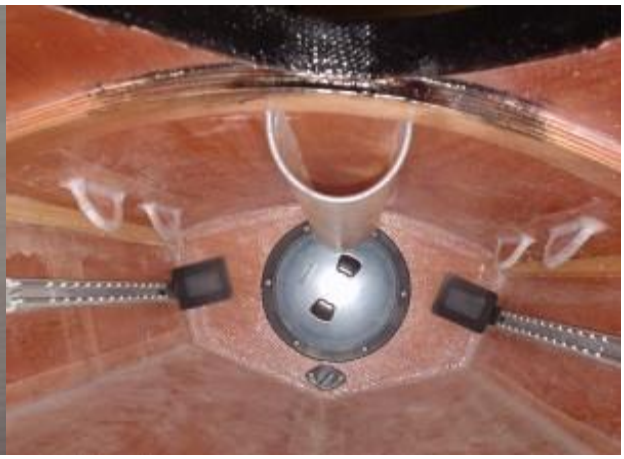


Another simple and low-cost solution from Alec Naneti in Romania uses two copper plumbing bends:



Another option is to create Maroske fittings:
We like the simplicity and elegance of Maroske deck fittings, but they are anything but simple to construct. Google “Maroske fittings”. It can be a struggle to pull the PVC tube out after the epoxy has set. To make it easy to extract the tube we first put the wire inside the PVC tube, then wrap the PVC tube with PTFE tape, and then insert the PVC tube into the fibre-glass sleeve. This entire assembly is then threaded through the deck. We use fiberglass tape over and under the sleeve to spread the load, and then thickened epoxy smoothed over the entire assembly. PTFE tape makes the PVC tubes easy to extract after removing the wire. We use a bottle cleaning brush to remove any remnants of PTFE that stick to the epoxy. One way of producing lightweight Maroske fittings is to bond in a semi-circle of 3 mm plywood, as in the left hand

photo below. This minimises the use of the heavy epoxy, and also produces an even curve, thereby facilitating the eventual withdrawal of the inner tube. When positioning the fittings near the cockpit, bear in mind the need for foot-room. Those in the right hand photo below are a little too close for comfort.

Key to open screw-in hatches

Screw-in hatches are light, and simple to install, and they don't interfere with the elegance of the kayak. However, they can be difficult to undo, particularly if subject to temperature changes. This is only exacerbated by having cold wet hands. We made a simple HDPE (bread or chopping board) key. One side opens the large hatches, and the other side opens the day hatch. We use both hands on the key, pushing and pulling, for maximum leverage. There are a couple of holes for a lanyard to attach it in the cockpit.




Weighing Epoxy

We use the excellent Smart Weigh digital scales to weigh the epoxy resin and hardener. The scales have an accuracy of 0.1g and a maximum capacity of 1 kg. They are equipped with a "tare" button that re-sets the scale to zero after the container is placed on the scale. This is a great feature. The scales cost £6 in the U.K. on eBay. One disadvantage of the scales is that they have a 60 second time-out.



The importance of sheer clamps

Some kit manufacturers omit sheer clamps, and rely on taping the seams between hull and deck. We

strongly advise against this with our lightweight hulls. The sheer clamps produce fair smooth curves and enable a very strong joint. We've spent too much time mending these joints on $3000 glass fibre kayaks where glass tape has been the sole method of joining the hull and deck.

Tethering Barton-style hatch covers

Peter M. from Cornwall, UK, used ½" 6g A4 s/s pan head screws and suitable washers to attach a nylon R-type cable clamp to the central moulding on the underside of the hatch cover. Knotted shock cord (3mm) joins the cover to another R-clamp inside of the hatch fitted to one of the bolts holding the hatch cover rim in place. This means that when the hatch cover turns, the central R-clamp turns freely and the shock cord is not wound up.



Finishing the edges of fiberglass cloth.

A clean finished edge is sometimes required on an area of glass cloth. This can be when, for example, just the floor of the cockpit is given an extra layer of glass cloth for abrasion resistance, or when a tapered extra keel strip is added. An elegant solution is to use dark coloured masking tape around the perimeter of the area, and laminate the glass over the desired area, overlapping slightly onto the coloured masking tape. After a few hours, when the cloth is set, but not hard, cut at the inner edge of the masking tape with a craft knife, and remove the surplus cloth and tape. The next day seal the cut line with resin.

Repairing a hole in a compartment.

Let's suppose you were rock-hopping, and holed the Vember's hull. You did, of course, have a buoyancy bag filling most of the compartment, enabling you to get home…… The challenge in repairing a hole in a bow, stern or day compartment is that there is no easy access to the inside to facilitate a simple patch repair, unless it is close to a hatch. In an emergency, just cover the hole in repair tape, but, when back in the workshop, here's how we do it: 1. Remove the damaged area by sawing round it with a jigsaw, or a hacksaw blade held in a rag. Make any corners well rounded, to reduce stress concentration. 2. Prepare a patch of 3mm marine ply (left over from your deck construction) to be the same shape as the hole, but about 25mm (1") larger all round its perimeter. Drill a hole of about 3mm diameter in the approximate centre of the plywood. 3. Coat both sides and the edge of the patch and the hole with epoxy resin, and let it set. 4. Lightly sand one side of the patch, and draw a conspicuous pencil line on this side about 1.5" (40mm) in from the perimeter, and all round it. Sand the first inch or so (25mm) of the inside of the compartment around the hole. 5. Pass about a foot (300mm) of 1/8th (3mm) cord through the hole in the patch, and put a bulky knot at the end of the cord on the un-sanded (inner) side, of the patch. Tie a one inch diameter loop in the cord on the outer side of the patch, and leave the excess cord intact. 6. Check that you can now "post" the patch through the hole in the hull, and pull it back by hand until it is firmly pulled into the correct position and orientation inside the compartment, as shown by an even view of the pencil line around the hole. The patch can be manipulated into position with one hand, while putting tension on the cord with the other. A temporary knob hot-glued into the centre of the outside of the patch can aid this procedure. Don't lose the patch inside the compartment! Tie a large tool to the other end of

the cord to make sure. Remove the patch and cord. 7. After the rehearsal, generously coat the outer inch or so of the perimeter of the sanded outer face of the patch with epoxy resin thickened to peanut butter consistency. The resin must hold its shape when the patch is vertical. 8. Delicately feed the patch through the hole, and pull the patch into its correct position and orientation. Slide a large screwdriver or piece of timber through the outer cord loop. Rotate the screwdriver to twist and thereby shorten the cord until the patch is pulled snugly against the inside of the hull, and an ooze of epoxy is visible. Maintain outward tension on the cord with one hand while rotating the screwdriver with the other. (This is known as a Spanish Windlass technique.) Tape the screwdriver to the outside of the hull when the desired tension is achieved, remove any excess epoxy, and leave the epoxy to set. 9. Remove the Spanish windlass system, cutting the cord and allowing the inner knot to fall into the compartment. 10. If the kayak has a painted finish, fill the resultant void with epoxy thickened with lightweight fairing compound, then sand and paint. If clear-finished, then use pieces of the hull stripping material to make a careful cosmetic repair. Then:
Mark an inch (2.5 cm) or so around the damaged area
Sand down to the wood within the marked area
Apply glass cloth over the marked area, overlapping onto the non-sanded bit by about a cm. (NB – make sure you've sanded the bit that's being overlapped to remove any varnish and key the epoxy without damaging the existing glass cloth)
Lightly sand the edge of the glass patch to feather in any hard lines
Epoxy over it once or twice so it's buried
Hand-sand the patched area so it has a matt finish.
If you varnish your boats, then cover the matt patch in varnish. If you don't use varnish then cover it in carefully-applied epoxy. NB – for a perfect finish, you could re-sand your entire kayak and completely varnish/epoxy her.

Black Keel strip.

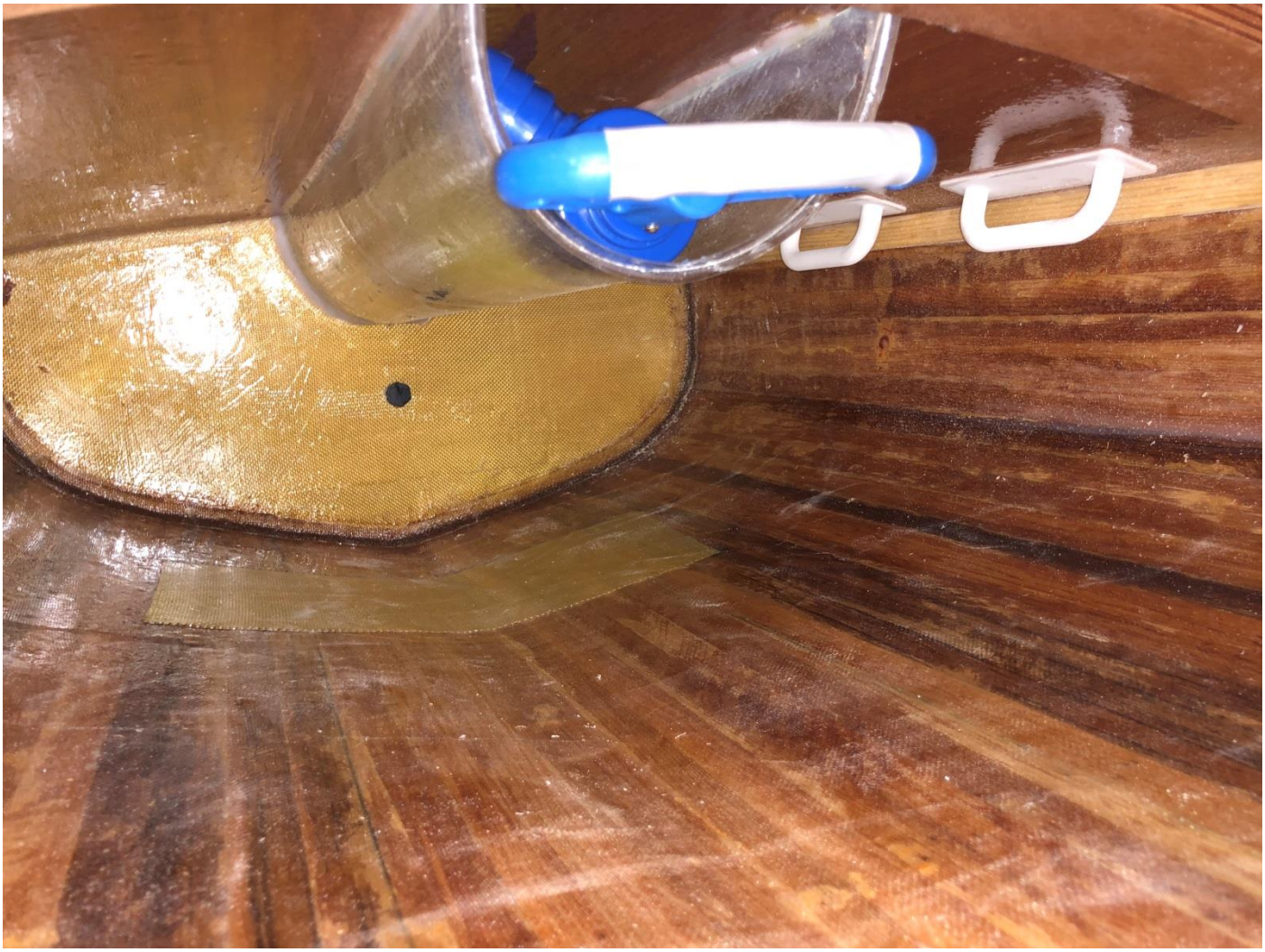
When the hull is left with a clear finish, it is difficult to spot when the keel has been scratched from contact with the beach or rocks. Water can then pass through the scratch and soak the plywood. This issue with a clear finish can be avoided by applying a coat of black pigmented epoxy (10% epoxy pigment) in a strip down the most vulnerable part of the keel, entirely below the water-line. If you taper the ends it can even look elegant. Good quality masking tape is worth the extra expense for this task. We use the blue 3M 50mmn (2 inch) wide tape.



Knee tube for pump

It seems a shame to spoil the lines of a beautiful kayak by having an ugly plastic pump on the deck,

however sensible and practical that may be. We install a knee tube under the deck. The knee tube construction is very basic. We use a length of 100mm diameter sewer plastic pipe as a mould, and laminate around the pipe two layers of whatever glass cloth we have handy, using either polyester or epoxy resin. We cut it to length and width when set, and glue some split plastic wire sheathing round the ends to protect feet, etc. We now angle the front of the tube for maximum foot clearance. We install the tube with the kayak upside down, using a few blobs of 5 minute epoxy, and later put a fillet of thickened resin down each side. We then install a piece of elastic shock cord across the mouth to stop the pump falling out. Note that you may have to shorten your pump. By removing an end cap this is usually possible. On some pumps we also temporarily remove the handle to shorten the rod.
Reed Chillcheater will supply spray skirts (decks) with bale tubes to enable access to the pump without removing the skirt, if you consider this necessary.



Thickening epoxy resin when used as a coating.

Occasionally one needs to place a coat of epoxy resin on a sloping surface, or on an exterior corner. One example is filling the weave on an existing keel strip, where the angular keel line makes the resin flow away from the apex of the upside down hull, which is exactly where one wants the thickest resin. Another example is filling the weave on parts of the cockpit rim and upstand. On all these we use 10% black pigmented epoxy resin with just sufficient West 406 Colloidal Silica filler powder added to make a “paint” that will still brush out smoothly, but will not run and sag as much as pure resin. This gives a very hard and shiny finish. We first mix the resin and catalyst very thoroughly, then add the pigment with much stirring, and then gradually add the powder until we obtain the desired consistency.

Ensuring the skeg box does not leak.

Two builders have reported water leaks from the top corners of the standard plywood skeg box. The plywood and inner solid wood framing of the box have simple joints at these points, so there is potential for leaks. We recommend special care to ensure the skeg box is water-tight. Do not use excessive clamping pressure which would exclude all epoxy adhesive from the joints. Apply glass fibre tape and resin over the outside perimeter of the box. Spoon a small amount of warm thin resin into the upside down box while it is on the bench, before the hole is drilled for the skeg wire fitting, and tip the box so as to run the resin into the joints. Leave the box upside down until the resin is set. Fill the box with water if you wish to check for any leaks. Taking extra care at this stage is less trouble than attempting to cure such a leak after the kayak has been completed and launched.

Installing a commercial hatch system



Instructions from Damian, in Plymouth, U.K:

Buy a large oval hatch rim. I found the 42/30 Kayaksport hatch fitted my standard Shrike perfectly.

1. *Make an approximately 4cm-wide oval ring out of 2 layers of 3mm ply so that the bottom lip of the hatch rim just locates inside it. (You do this in the same way as you manufacture the cockpit rim for a Shrike.)*
2. *Decide where you want the hatch to go on the front deck. Then draw the outside edge of the plywood ring onto the deck.* ***(NB: when you sit the ring on your curved deck, the left and right sides will be dangling a few inches above the deck. Trace round the ring keeping it horizontal, and your pencil vertical to get the right shape on your deck.)*** *Draw a line approximately 4cm outside the rim outline and cut it out, e.g. with a jig-saw. (NB – I am saying 4cm here, but it's your choice: whatever distance you choose will determine the angle of the recess from the deck down to the hatch rim. You'll probably want to have different angles at different points on the rim to make it look nice – that's fine.)*

3. *Put the plywood oval ring into the hole in your deck. On a standard Shrike, the edges of a large hatch rim should be sitting on the port and starboard shear clamps. (If not, you'll need to support it while you do the remainder of the work.) Use sandpaper/files/plane/etc. to make sure everything is neat and symmetrical and that you're happy with the angle of the slope down from the deck to the hatch rim. Then use superglue and accelerant to stick the plywood ring in position on the shear clamps.*
4. *Cut out lots of pieces of plywood to fill the gap between the oval ring and the deck. This gap will start at zero by the shear-clamps and get bigger as the deck rises. I used about 20 pieces in total. Shape them with a plane and sandpaper. Don't worry if it looks a bit rough-and ready: you'll fill gaps with epoxy and sand them smooth later. Bevel the edges of these pieces so they butt up against the deck and the cockpit rim.*
5. *Use super glue and accelerant to fix the gap-fillers in place.*
6. *Apply an epoxy fillet around the underside of the top joint (i.e. where the gap-fillers meet the deck) and fill any gaps between the gap-fillers.* ***NB: if you have any big holes, then cover them with masking tape from above, turn the boat upside down and fill them with thickened epoxy from underneath. Remove the tape when the epoxy has gone off and sand it smooth.***
7. *Put a layer of fibre-glass over this fillet – either glass tape (cutting the edges to allow it to go round corners) or cut glass cloth to shape.*
8. *Put a fillet on top of the bottom joint (i.e. where the gap-fillers meet the plywood ring) and use thickened epoxy to fill in any remaining gaps.*
9. *Once this fillet has hardened, smooth off the underside of this joint with sandpaper and then apply fibreglass to it – either glass tape or glass cloth.*
10. *Finally, sand the top surface of both top joints smooth and cover both in glass – either tape or cloth. I used 160 gram 'twill' cloth which will lay flat over curving surfaces, so I was able to easily cover both joints with one piece. At this point, make sure you've painted any exposed plywood with epoxy.*
11. *When the epoxy has just gone off, cut the corners off the glass with a Stanley (craft) knife, apply more layers of epoxy (to bury the fibreglass weave) and then sand it smooth and flush with the surrounding deck. Then finish it however you prefer – varnish, epoxy or paint.*
12. *Fit your hatch rim to the plywood ring using adhesive. I used a silicone sealant, and then drilled/bolted it in place.*
13. *Buy a cover to fit your hatch rim. Tie it on to your deck-lines (so it doesn't fly away if it comes off while you're driving.) Fit it. Job done.*

Low profile skeg box

The skeg system shown in the Shrike plans works well with full-sized Shrikes and also with Shrike LVs where the plans are all scaled down, including the size of the skeg box and skeg. However, if only the height of the decks is reduced, as in the Shrike-R, there may be insufficient space between the top of the keel and the underside of the stern deck to accommodate the system. The deck may foul the compression fitting on top of the skeg box.

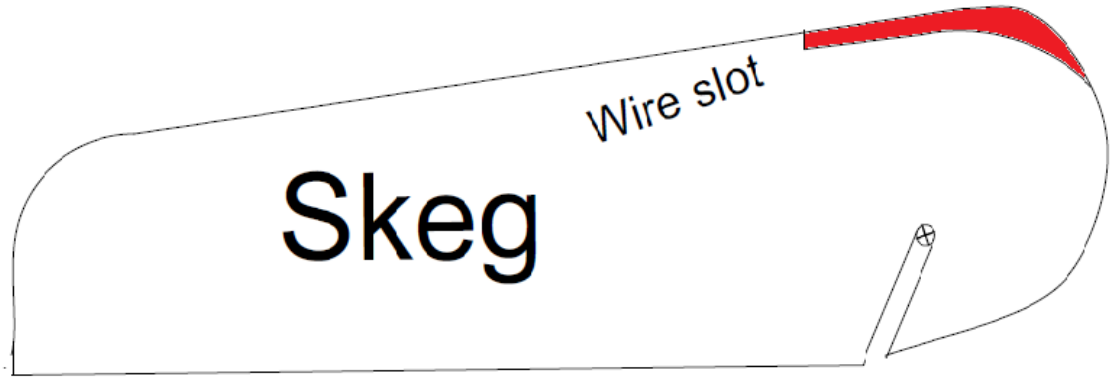
This can be avoided by installing the compression fitting in the front of the skeg box as shown in the photo below. The following instructions should be read in conjunction with the accompanying photos, and the appropriate section of the Build Manual in the download:

1. Print the paper plans for the skeg box and skeg at 90%, thereby reducing the deck clearance height by about 13mm.

2. Increase the width of the inside of the box to 15mm by using 15 x 12mm cedar internal framing. The overall dimensions of the 90% box are 346 x 122mm.



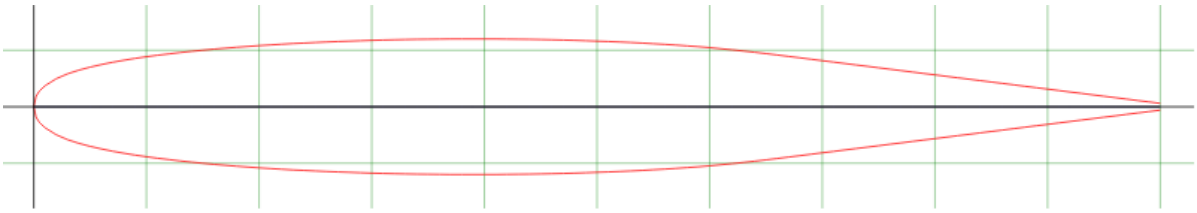
3. Make a 9mm thick skeg blade from three layers of 3mm plywood, instead of the standard two-layer 6mm thick blade. Before gluing the three layers together, the top curve of the middle layer is profiled to accept the 3mm stainless wire cable in a groove, where it is later epoxy bonded. Cut the inner layer of plywood to accept the wire by reducing the radius of the curved end to a circle centred on the pivot point, and meeting the bottom of the straight slot for the wire, as shown in the area coloured red, which should be removed:



After the three layers are bonded together, taking care to remove excess glue, shape the groove with a round file to accept and guide the free wire over the range shown in the photos, and epoxy glue the wire into just the straight section of the groove:



4. Round the leading edge of the blade and taper the trailing edge in accordance with the NACA 0012 profile (thickness = 12% of width), a good shape for optimizing lift at normal kayak speeds: (C.A Marchaj, “Sailing theory and practice”, 1964, pp 281 – 284 & airfoiltools.com)

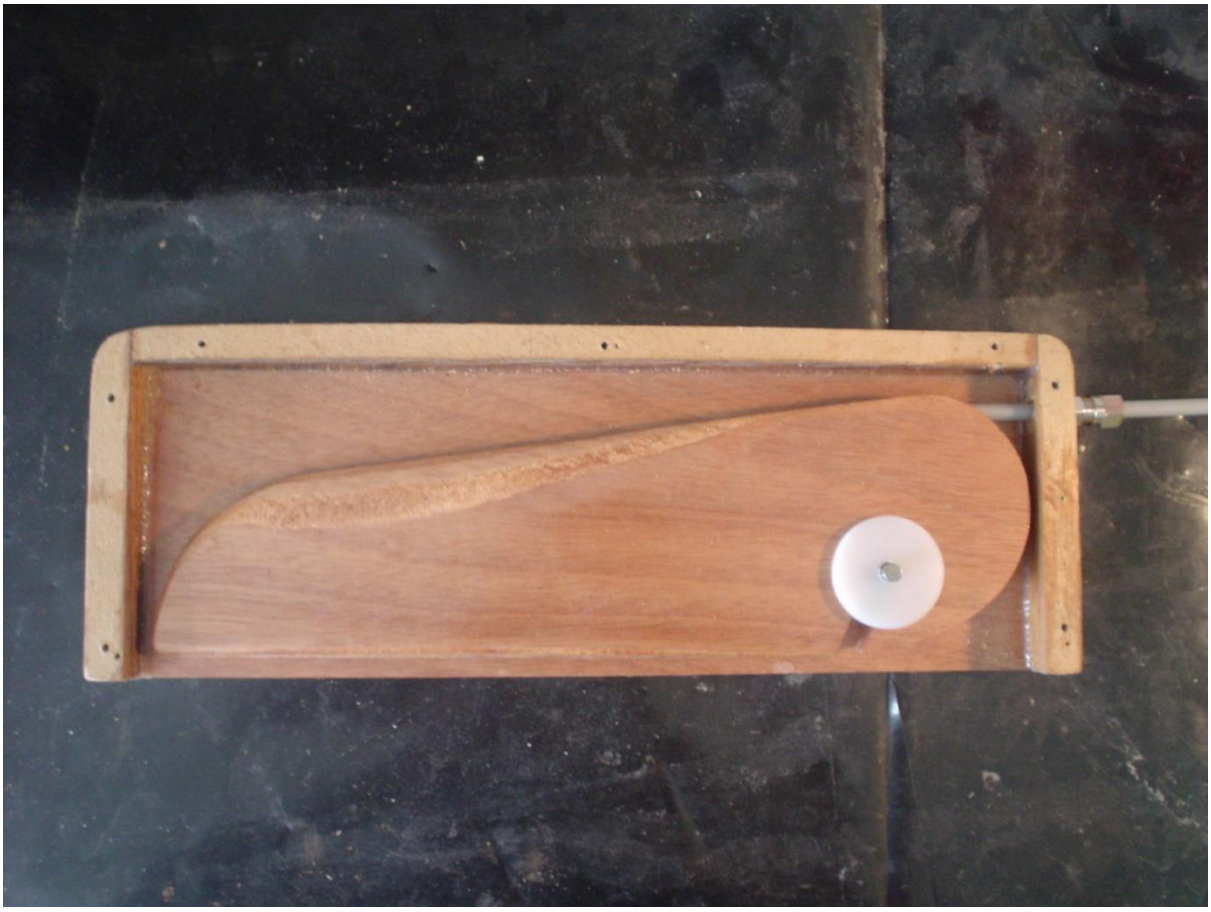


Give the skeg and box a coat of epoxy, sanded hard, and then a second coat, to ensure smooth contact surfaces.

5. Fit the compression fitting horizontally, with its centre line 95mm above the bottom of the box.

6. When the outer 6mm O.D/ 4mm I.D plastic tube is inserted into the compression fitting, the plastic outer tube should extend into the skeg box by 25mm. With the compression fitting we used, this gave a measurement of 60mm from the outside of the compression fitting to the end of the tube. A pen mark or piece of tape 60mm from the end of the tube helps to position the tube correctly.

7. Fit nylon spacer washers as necessary on the pivot, and complete the box.



The box, when slotted through the keel, will require a height clearance of 119mm from the top of the keel to the underside of the stern deck. A skeg system to this design operates very smoothly, but disadvantages of this design are that the compression fitting and tube are more vulnerable to damage from stowed equipment in the stern compartment, and the available storage volume is reduced.

Adjusting the top of the foot bulkhead when the masik curve has been lowered.

If you decide to reduce the curvature of the masik to lower the front of the cockpit you will also need to reduce the height of the foot bulkhead.

The simplest way to do this is to place the foot bulkhead plywood against the masik and trace the same curve onto the top of the foot bulkhead. Different plywood will bend in different ways, so if you wish to be more precise you can create the curve at the top of the top of the foot bulkhead using the same equation used to create a bespoke masik, with a new beam (L) and new height (D) above the gunwales. The beam (L) is easy to measure at the foot bulkhead, but the height (D) is more complicated: Without the foot bulkhead in position, fit the foredeck plywood in its final place without any glue, but with enough straps to make the plywood buckle in its final position about 900 mm from the bow. Then measure the vertical distance (E) from the top of the keel to the underside of the deck plywood at the foot bulkhead position, inside the hull. We use a steel tape measure with a lock to do this. The body of the tape is usually marked with its width, so you can jam the tape in position and thus obtain an accurate measurement. (E)

Now remove the deck, place a straight edge across the hull at the foot bulkhead position, and measure the vertical distance (F) from the keel to the underside of the straight edge. Subtract this distance from the previous measurement, and you have the height (H = E - F) to put into the equation.

Whichever method you use, fit the foot bulkhead temporarily in place with hot glue or drops of CA glue, fit the deck with straps, and mark with a pencil any places where the foot bulkhead top is too high. Take off the deck, adjust the top of the foot bulkhead with a plane, and repeat until you are satisfied. Then add the strips to make the top of the foot bulkhead thicker so as to make a stronger joint, as shown in the Build Manual.

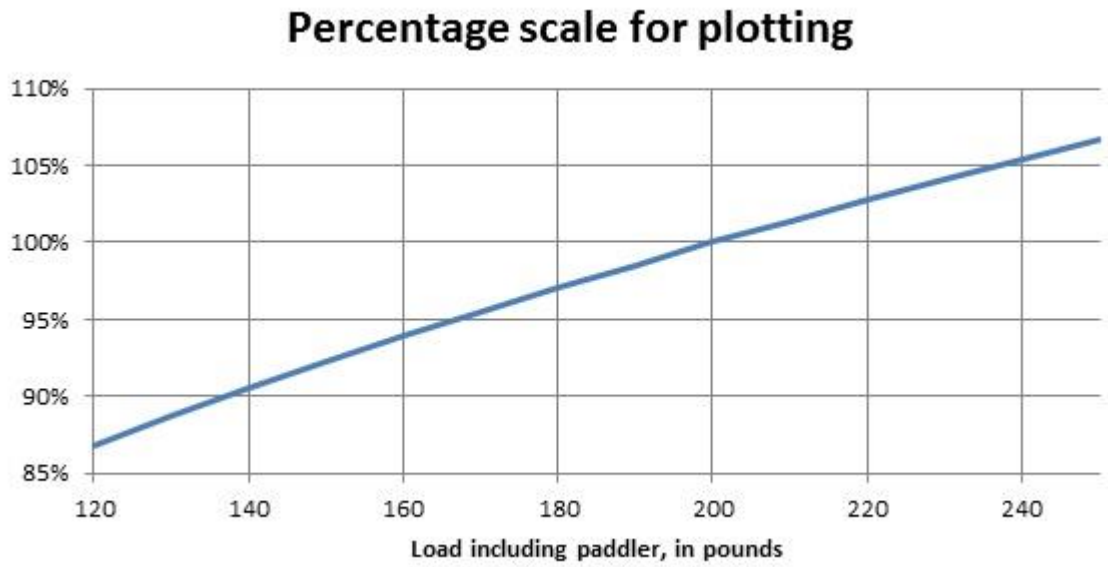
Appendix 4: Frequently asked questions

How do I decide which model and variations to choose?

Vember is designed as a responsive rough water day-paddling kayak, and Vembex is the 10% lengthened version for extra load capacity for expedition use. One can also vary the scale of the plans to suit your bodyweight, as determined by the graph below under "Will Vember work for my weight?" Finally, select your desired height for leg clearance at the front of the cockpit, between the keel and bottom of the cockpit rim. Experience in other kayaks will guide you in this.

Will Vember work for my weight?

As described in the free downloadable Build Manual, the height of the gunwales can be varied to suit differing paddler and equipment loads. However, the scale of the plans can also be changed during printing, so that every dimension of the kayak is adjusted by the same percentage. Below is the graph we use to relate the printer/plotter percentage scale to various loads of paddler plus equipment, in pounds. (For Vembex you can add 64 pounds to the load on the x-axis, as shown in the photos in the Build Manual)



Because of the length limitations in the pdf specification, the pdf is downloaded at 50% scale, so percentages taken from the above graph must be doubled. For example, a load of 137 pounds (62.1 kg) suggests a 90% plot, doubled to 180% if the pdf is downloaded at 50%. Feel free to experiment with your planned design. Bear in mind when scaling the design that all linear dimensions must be changed to the same scale. That includes all the measurements relative to the datum line, such as paddler seat centre, position of temporary forms (exterior and interior), and position of bulkheads

How do I know that my plans are printed accurately?

The plans include two lines, each 100 mm long, one along and one across the print-outs. These lengths should be checked with a steel rule to ensure accuracy. Our comfort level is plus or minus 0.5 mm.

Should I use cove and bead or rectangular section timber? As mentioned in the Build Manual, using pre-shaped cove and bead timber is a fast and simple way to produce the hull, and the cove retains the glue for much of the construction. As Vember was our first attempt at strip-building we decided to use this method. However, we now prefer to use rectangular section 19 mm x 6 mm or 4.5 mm timber without milling for cove and bead. Each strip is hand-planed on one edge as it is test-fitted into place before gluing. This will take a little more time and patience, but it can be a lower cost and lower weight option.

Should I use stapled or clamped construction?

We chose to use clamps as we didn't want the staple holes and the tedious task of removing staples. However, there are simple ways to deal with these issues. Moreover, clamping obstructs the next strip position, so we only fitted two pairs of strips each day (one pair in the morning and one pair in the evening) Using staples to enable continuous stripping may be desirable for professionals, but, as beginners, we were using half-length strips to make the fitting of the tapered ends less critical. This then requires at least a butt-joint part-way along the hull. The simplest way to ensure this joint is fair is to clamp it, again obstructing placing of the next strip. There are ways around this, perhaps using super-glue or hot glue, but we wanted a simple fool-proof method, and clamping worked for us. Stripping the hull is very enjoyable, exciting and rewarding.

How strong is the hull?

The timber core, surrounded by inner and outer layers of epoxy and glass cloth forms a composite structure of remarkable strength. We removed a strip of Vember's hull for the skeg slot, and we were unable to snap it in our hands. Here's a report from Damian, who thought he'd wrecked his Vembex:

"*The composite construction is incredibly strong. On a recent outing, I was carrying my Vembex across a rocky shore and completely failed to spot a knee-height boulder. When I tripped over it, I flew forward and ended up crashing down on some sharp granite rocks with my full bodyweight on top of the boat. If you wanted to smash a boat in one manoeuvre, this is how you'd go about it - a plywood boat would have been garden-waste-sack-material, and I reckon it could have holed a fibreglass boat. However, the composite sandwich construction laughed it off - the only damage was a few patches with little white 'brickwork' patterns, where the fibreglass had delaminated along joins between wood strips. It paddled back fine, and was completely dry.*

Repairing this kind of damage is actually very easy - it's just a matter of sanding away any damaged fibreglass, and replacing it with a fibreglass patch which overlaps the existing glass. By 'feathering in' this patch with sandpaper and then re-coating with epoxy/varnish, the repair becomes invisible."

Does Vember need the skeg?

Whether or not a skeg is necessary very much depends on the area you paddle, what are the wind and sea conditions in which you paddle, and your own physical state. For example, Nick has a chronically damaged right shoulder. He can't afford to edge and sweep for long periods to counteract weather-cocking. The worst conditions for weather-cocking are flat sea and strong winds from the quarter. If you always aim to paddle in light winds, and you are fit and strong, then you may never need to deploy a skeg in a well-balanced kayak. If you are in doubt, you can build a Vember without a skeg, but with a stern hatch and the stiffening structure for a skeg. You could then retro-fit a skeg if you decide one is desirable, although this is more awkward than fitting one during the initial construction. In the end though, we have not yet paddled a sea kayak that, ultimately, and in certain conditions, did not benefit from the deployment of a skeg. This becomes particularly relevant on long trips in varying wind condition, when prolonged edging and sweeping can become exhausting.

Can I omit sheer clamps, and rely on glass tape inside and outside the deck to hull joint?

We strongly advise against this. The sheer clamps produce fair smooth curves and enable a very strong joint. We've spent too much time mending these joints on $3000 glass fibre kayaks where glass tape has been the sole method of joining the hull and deck.

Can I use cheaper plywood for the deck?

The recommended BSS 1088 3mm okoume plywood is usually of superb quality. It is consistent in that we have experienced no voids in the inner ply, and it bends evenly without splitting. We have experimented with cheaper plywood, with conspicuous lack of success.

How many hours does it take to build a Vember?

Obviously this depends on one's experience but an estimate, including some thinking and reading time, would be 150 hours for a first-time strip builder.

Why the name "Vember" and why the name "CNC kayaks"?

Vember, like Shrike, is the name of an extreme rock climb on Clogwyn d'ur Arddu, a cliff on the north face of Yr Wyddfa, the highest peak in Wales. Vember is also a girl's name in Wales. More particularly, "a vember" has recently come to mean "an adrenaline rush".

We like to play with words. CNC is, of course, widely used as an abbreviation for "Computer Numerically Controlled", that describes the method by which computers take over from humans in much of industry. We design our kayaks by hand and eye, rather than be constrained by the limitations built into any controlling software. We only use a computer to receive the manual measurements of the completed kayak, and to output these measurements to plans. So, CNC is used tongue-in-cheek, knowing that it is likely to be misinterpreted. The truth is that **C**hristopher & **N**ick **C**rowhurst is the reason for the label.